

Quantum Technologies for Autonomous Robotics: A Systematic Survey of Methods, Evidence, and Deployment Constraints

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Quantum computing promises speed-ups for optimization, search, sampling, and learning, and these are exactly the computations that hold autonomous robots back in large, uncertain, and changing environments. Yet the claims made for “quantum robotics” range from asymptotic complexity results to small simulations and a handful of hardware runs, and it is hard to tell which of them a robot could use today. This article presents a systematic survey, reported following the PRISMA 2020 statement where it applies, of 72 primary studies that apply quantum or quantum-inspired methods to robotic functions, set against 107 foundational works on quantum algorithms, hardware, cognition, and classical robotics and six related quantum computer-vision studies. We organize the field by robotic function (optimization, planning, learning, sensing and communication, and decision-making) and grade every study on a six-level evidence hierarchy that runs from theoretical analysis to end-to-end deployment. For each study we also record whether its quantum component ran on quantum hardware, in simulation, or not at all, and we keep genuinely quantum results apart from quantum-inspired classical ones. The picture that emerges is sobering. No quantum computational method has yet shown an end-to-end advantage on a physical robot. Quantum annealing has routed fleets of up to 1,000 simulated vehicles after decomposition, but classical and quantum-inspired solvers match or beat it whenever the two are compared directly. Quantum reinforcement learning agents reach comparable returns in simulated robot tasks with one to two orders of magnitude fewer trainable parameters, but so far this has not meant shorter training, fewer samples, or better policies. Cloud latency, noise, and trainability still keep quantum processors out of real-time control loops. Only quantum sensing and quantum communication have been fielded on moving platforms, and the single field result showing a navigation advantage still awaits independent replication. We identify quantum-probability models of human judgment as a hardware-light opportunity for human-robot interaction, propose reporting standards for quantum-robotics experiments, and derive a research agenda from the taxonomy.

CCS Concepts: • **General and reference** → **Surveys and overviews**; • **Computer systems organization** → **Robotics**; • **Hardware** → **Quantum computation**; • **Theory of computation** → *Quantum computation theory*; • **Computing methodologies** → *Planning and scheduling*; *Machine learning*.

Additional Key Words and Phrases: Quantum robotics, quantum annealing, QAOA, variational quantum algorithms, quantum machine learning, quantum reinforcement learning, quantum sensing, hybrid quantum-classical systems, quantum cognition, NISQ, systematic literature review

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1 INTRODUCTION

1.1 Motivation

An autonomous robot solves hard computational problems over and over, and it rarely has much time to do so. It has to estimate its state from noisy sensors, plan motions that avoid collisions, share tasks with other robots, and learn behaviors from limited experience. Probabilistic estimation, sampling-based planning, and learning have taken robotics a long way [153, 166], but many of the underlying problems are still expensive. Multi-robot routing and task allocation are NP-hard combinatorial problems, particle-filter localization and belief-space planning scale with the size of the state space, and reinforcement learning on physical robots is notoriously sample-inefficient [93].

This is where quantum computing has drawn attention, because on paper it offers new tools for exactly these problems. Grover's algorithm gives a quadratic speed-up for unstructured search [69], amplitude estimation accelerates Monte Carlo sampling quadratically [115], variational algorithms such as the Quantum Approximate Optimization Algorithm (QAOA) target combinatorial optimization [52], and parameterized quantum circuits have been proposed as compact function approximators for learning [26]. Quantum sensors, for their part, are starting to deliver measurable gains in navigation-grade inertial and magnetic sensing [39].

Robots, however, come with constraints that papers on quantum algorithms seldom face. Decisions are due within milliseconds to seconds. Size, weight, and power on board are limited. Problems change while they are being solved and are often continuous, and safety demands answers that are reliable, not merely probable. Today's noisy intermediate-scale quantum (NISQ) devices are cryogenic, remotely accessed, noisy, and small [135]. We therefore set aside the question of whether quantum computing could help robotics in principle. We ask instead what the published evidence shows: where it helps, by how much, and when. To answer that, we follow the field from Benioff's description of a quantum robot in 1998, through the annealer and variational experiments of the NISQ era, to the first attempts in 2025 and 2026 to put a quantum processor inside a robot's control loop. Along the way we grade each step by the strength of its evidence.

1.2 Scope and Terminology

We use *quantum robotics* for the application of quantum computation, quantum sensing, quantum communication, and quantum-probability modeling to the functions of autonomous robots. Three distinctions recur throughout the survey. First, a method is *genuinely quantum* if it requires quantum hardware, or its classical simulation, to run, and *quantum-inspired* if it is a classical algorithm that borrows quantum concepts such as amplitudes or superposition-like representations. Second, a method is *NISQ-compatible* if it can run on today's noisy processors without error correction, and *fault-tolerant* if its advantage depends on error-corrected logical qubits. Third, we separate *quantum computation* for robots from *quantum sensing and communication* on robots, because the two are at very different levels of maturity. Quantum computation is our main subject. We include quantum sensing, quantum communication, and quantum-probability models as well, because these are the branches in which quantum technology has already reached robots, or comes closest to doing so, and they show what a working quantum component on a robot actually looks like. We do not cover robots that manipulate quantum systems as their task (for example, laboratory automation of quantum experiments) or the use of classical robotics to build quantum hardware.

1.3 Related Surveys and Positioning

Earlier surveys have mapped possible intersections between quantum technologies and robotics. Tandon et al. [160] wrote the first book-length survey, organized around how quantum methods could help a robot sense, plan, learn, and act. Petschnigg et al. [132] published a position paper at

ICRA that mapped quantum algorithms onto the sense-think-act loop and framed robots as future users of quantum cloud services. Yan et al. [178] reviewed quantum-robot architectures, perception, and robot-environment interaction, including quantum-inspired evolutionary algorithms, quantum affective computing, and quantum cryptography for robot communication. Since 2024 the pace has picked up. Nigatu et al. [121] reviewed 107 contributions on quantum machine learning for robotics in five application domains. Udekwe [168] recast robotic tasks as optimization, search, sampling, and linear-algebra problems and discussed hardware and latency constraints. Shorter surveys have covered quantum-based robot control [54] and quantum computing applications in robotics [72]. Broader reviews provide taxonomies of quantum computing as a whole [63], connect quantum computing with artificial intelligence [59], and discuss its role in automating science and engineering [82]. Surveys of the underlying methods are not specific to robotics. They cover quantum reinforcement learning [84, 113] and quantum machine learning [172], including a recent survey in this journal [144]. The closest neighbor is a survey of quantum computing for intelligent transportation [184], which covers vehicle routing and autonomous driving but stops short of robots.

Table 1 compares these robotics surveys with the present one. Most of them are descriptive. They rarely grade the evidence behind a claim, so asymptotic speed-ups, noiseless simulations, and hardware experiments end up side by side. Few separate genuinely quantum methods from quantum-inspired classical algorithms, even though the latter run on ordinary hardware and account for many of the robotics results that have been validated physically. And because most of them predate the experimental work of the NISQ era, or give it little weight, they say little about quantum reinforcement learning on robot tasks, QUBO-based fleet control on quantum annealers, head-to-head solver comparisons, dequantization results, error-mitigation costs, and measured cloud latencies.

The two closest recent reviews differ from ours in kind, not in size. Nigatu et al. [121] cover more works (107) but restrict themselves to quantum machine learning and do not grade evidence, and Udekwe [168] organize the field by computational problem class and grade evidence only partially. This survey covers fewer machine-learning papers than Nigatu et al. [121] but spans five robotic functions and four technologies, grades every study on the same evidence scale, records where its quantum component ran, and tests the claims against the latency budgets of real robots. Its corpus is young: 34 of the 72 primary studies appeared in 2025 or 2026, and 16 were available only as preprints when they were graded; Section 3.7 discusses what this means for the conclusions.

1.4 Contributions

This survey makes five contributions.

- (1) A systematic review, reported following PRISMA 2020 where it applies, of 72 primary studies on quantum and quantum-inspired methods for robotics, published between 1998 and 2026, set against 107 foundational works and six related quantum computer-vision studies (Section 3).
- (2) A robotics-function taxonomy that links each robotic function to the quantum methods applied to it and to representative tasks, together with a six-level evidence hierarchy, a quantum-execution tag, and five further classification dimensions that are applied to every study (Section 3.6, Figure 3).
- (3) A critical, method-by-method synthesis of quantum optimization, search, learning, sensing and communication, hybrid architectures, and quantum cognition for robots, in which every result is reported with its evidence level, baseline, and principal limitation (Sections 4 to 10).

Table 1. Comparison of this survey with earlier surveys of quantum computing for robotics. ✓ = addressed systematically; (✓) = addressed partially or descriptively; – = not addressed. Evid. = evidence; Q vs. QI = genuinely quantum results separated from quantum-inspired ones; HW = hardware; comm. = communication; QP = quantum probability. The assessment is ours and is based on the content of each survey.

Survey	Type	Focus	Evid. grading	Q vs. QI	HW and latency	Sensing, comm.	QP cognition
Tandon et al. [160]	Book	Sense, plan, learn, act; quantum MDPs and POMDPs	–	–	(✓)	✓	–
Petschnigg et al. [132]	Position paper	Sense-think-act loop; quantum AI, perception, optimization	–	–	(✓)	(✓)	–
Yan et al. [178]	Narrative review	Architectures, perception, robot-environment interaction	–	(✓)	–	✓	(✓)
Haldorai [72]	Narrative review	Quantum computing principles and robotic applications	–	–	(✓)	–	–
Udekwe [168]	Systematic review	Robotic tasks recast as optimization, search, sampling, linear algebra	(✓)	(✓)	✓	(✓)	–
Fazilat and Zioui [54]	Short review	Quantum-based control strategies	–	–	–	–	–
Nigatu et al. [121]	Review	Quantum machine learning for robotics (107 works)	–	–	(✓)	–	–
This survey	Systematic review (PRISMA-informed)	Five robotic functions; 72 primary studies, 185 references	✓	✓	✓	✓	✓

- (4) A cross-cutting analysis that separates theoretical, simulated, hardware, and end-to-end claims of quantum advantage and quantifies the latency gap between robot control loops and quantum hardware access, together with a worked fault-tolerant resource estimate for Grover-based localization (Section 11).
- (5) A research agenda derived from the taxonomy, together with a reporting checklist for future quantum-robotics experiments (Section 12).

1.5 Organization

Section 2 introduces the quantum-computing concepts used in the survey, the computational kernels of autonomous robots, and the history of the field. Section 3 describes the survey methodology and the classification scheme. Sections 4 to 10 review the literature by method family: optimization (Section 4), search and planning (Section 5), learning (Section 6), sensing and communication (Section 7), hybrid architectures and hardware constraints (Section 8), quantum cognition (Section 9), and emerging directions (Section 10). Section 11 synthesizes the evidence across methods, Section 12 sets out open challenges and a research agenda, and Section 13 concludes.

2 BACKGROUND

2.1 Quantum Computing Essentials

A qubit is a two-level quantum system whose state is a unit vector $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$. A register of n qubits is described by 2^n complex amplitudes; gate-based quantum computers transform it with unitary gates and read it out by measurement, which returns a classical bit string with probability given by the squared magnitude of its amplitude [120]. Quantum algorithms obtain speed-ups by arranging for the amplitudes of wrong answers to cancel through interference, not by “trying all answers in parallel”. Two further consequences matter for robotics. Measurement destroys the superposition, so expectation values have to be estimated from many repeated runs, called *shots*. And loading classical data, such as a sensor reading, into amplitudes costs circuit depth, which can eat up the speed-up [1].

Two hardware paradigms appear in the robotics literature. *Gate-based* processors, built from superconducting circuits, trapped ions, or photons, run arbitrary circuits [92]. *Quantum annealers* implement a single algorithm: they evolve slowly from an easy initial Hamiltonian to a problem Hamiltonian whose ground state encodes the solution of an Ising or quadratic unconstrained binary optimization (QUBO) problem [81, 104],

$$\min_{x \in \{0,1\}^n} x^\top Q x = \min_{x \in \{0,1\}^n} \sum_{i \leq j} Q_{ij} x_i x_j, \quad (1)$$

where constraints such as “each robot takes exactly one route” are added as quadratic penalty terms. On gate-based hardware, the same QUBO can be attacked with QAOA, which prepares $|\gamma, \beta\rangle = \prod_{k=1}^p e^{-i\beta_k H_M} e^{-i\gamma_k H_C} |+\rangle^{\otimes n}$, alternating p layers of a cost Hamiltonian H_C and a mixing Hamiltonian H_M , with angles tuned by a classical optimizer [52]. QAOA belongs to the family of *variational quantum algorithms* (VQAs), together with the Variational Quantum Eigensolver (VQE) and the parameterized circuits of quantum machine learning. All of them run a short circuit on the quantum processor and leave the parameter updates to a classical computer [15, 26, 111, 131].

Current processors belong to the NISQ regime: tens to a few thousand physical qubits, gate errors of roughly 10^{-3} to 10^{-2} , and no error correction [135]. Error mitigation can improve expectation values at a sampling cost that grows exponentially with circuit size [25, 85, 163], whereas fault-tolerant computing encodes each logical qubit in many physical ones [58, 165]. This frontier has moved fast. With error mitigation alone, an IBM processor produced reliable expectation values for 127-qubit circuits that brute-force classical simulation cannot reach [89], although approximate classical methods soon reproduced the same observables faster than the experiment [12]. A neutral-atom processor ran algorithms on up to 48 logical qubits [18], and Google’s Willow processor showed logical errors falling as the surface code grew, which places it below the error-correction threshold [67]. High-rate low-density parity-check codes may cut the physical-qubit overhead of fault tolerance by about an order of magnitude [21]. The algorithms with provable speed-ups relevant to robotics (Grover search, amplitude estimation, and quantum linear-system solvers [74]) require fault tolerance; the NISQ-compatible algorithms (QAOA, VQE, variational learning, and annealing) are heuristics whose advantage, if any, must be shown empirically [116].

2.2 Computational Kernels of Autonomous Robots

The robotics problems that quantum methods have been aimed at fall into five kernels. *Combinatorial optimization* covers task allocation, multi-robot routing, scheduling of automated guided vehicles (AGVs), and assembly-line balancing. These problems are usually NP-hard and are solved classically with mixed-integer programming or metaheuristics. *Search and planning* covers motion and path planning, for which sampling-based planners such as RRT [98] and the asymptotically optimal RRT*

and PRM* [86] are the classical baselines, together with integrated task-and-motion planning [60]. *Estimation* covers localization and mapping, where Monte Carlo localization [40] and particle-filter SLAM scale with the number of hypotheses. *Learning* covers supervised perception and reinforcement learning (RL) [93, 158]. *Decision-making under uncertainty* covers partially observable Markov decision processes (POMDPs) [83, 95] and models of the humans with whom robots interact. These five kernels map onto the five robotic functions of our taxonomy (Section 3.6).

2.3 Historical Evolution

The history of the field is short but not as short as it looks. Figure 1 traces it in four eras, with milestones in quantum computing on one side and in robotics on the other. In the *theoretical era* (1982–1993), Feynman [56] argued that classical computers cannot efficiently simulate quantum systems, Deutsch [41] formalized the universal quantum computer, and Deutsch and Jozsa [42] gave the first exponential separation between quantum and deterministic classical computation. In the *algorithmic era* (1994–2005), Shor’s factoring algorithm [152], Grover’s search [69], and amplitude amplification and estimation [20] showed that quantum computers could outperform classical ones on practically important tasks. Robotics entered the picture as theory: Benioff [13] introduced the *quantum robot*, a mobile quantum system carrying an onboard quantum computer and interacting with its environment, and later showed that such a robot gains nothing from Grover search when it must move through the region it searches [14], while quantum-inspired evolutionary algorithms brought quantum ideas to classical optimization [73].

In the *early-hardware era* (2006–2015), Dong et al. [47] turned Benioff’s concept into an engineering architecture of quantum computing units, controllers, and sensors, and proposed Grover-based search and quantum reinforcement learning for it [46]. On the computing side, the HHL algorithm promised exponential speed-ups for linear systems [74], D-Wave demonstrated programmable quantum annealing [81], and VQE [131] and QAOA [52] introduced the variational approach. Interestingly, the first robotics results of this era did not use quantum computers at all, but classical hardware and sensors: quantum-inspired reinforcement learning drove a physical robot [45], an atom-interferometric accelerometer flew on an aircraft [61], and planning problems were first mapped to a quantum annealer [143]. In the *NISQ era* (2016–present), public cloud access opened gate-based processors to robotics groups, Preskill [135] named the regime of noisy 50–100-qubit devices, and Google’s Sycamore processor performed a random-circuit sampling task beyond the reach of classical simulation at the time [5]. Since 2023 the NISQ era has begun to give way. Error-mitigated circuits on 127 qubits reached beyond brute-force simulation [89], though approximate classical methods later matched them [12], logical qubits were operated in numbers [18] and below the error-correction threshold [67], lower-overhead codes appeared [21], and a quantum annealer simulated spin-glass dynamics that its authors argue are beyond classical reach, although that claim is still contested [90]. Over the same years quantum robotics moved from concepts to experiments. Annealers routed vehicles and robot fleets [34, 118, 123, 136], gate-based circuits were applied to motion planning [30], swarms [29], localization [4], kinematics [126], and navigation learning [77]. Drones carried entanglement and quantum keys [101, 167], and quantum magnetometers navigated aircraft without GNSS [117]. Most recently, a quantum policy controlled a physical system [157] and a QPU was placed inside a control loop [119].

3 SURVEY METHODOLOGY

This survey follows the PRISMA 2020 reporting guideline [127] and the systematic-review procedure of Kitchenham and Charters [91], which divides a review into planning, conduct, and reporting stages. We report the PRISMA items that apply to a review of a fast-moving computing literature, and we say plainly in Section 3.4 which record counts we cannot report.

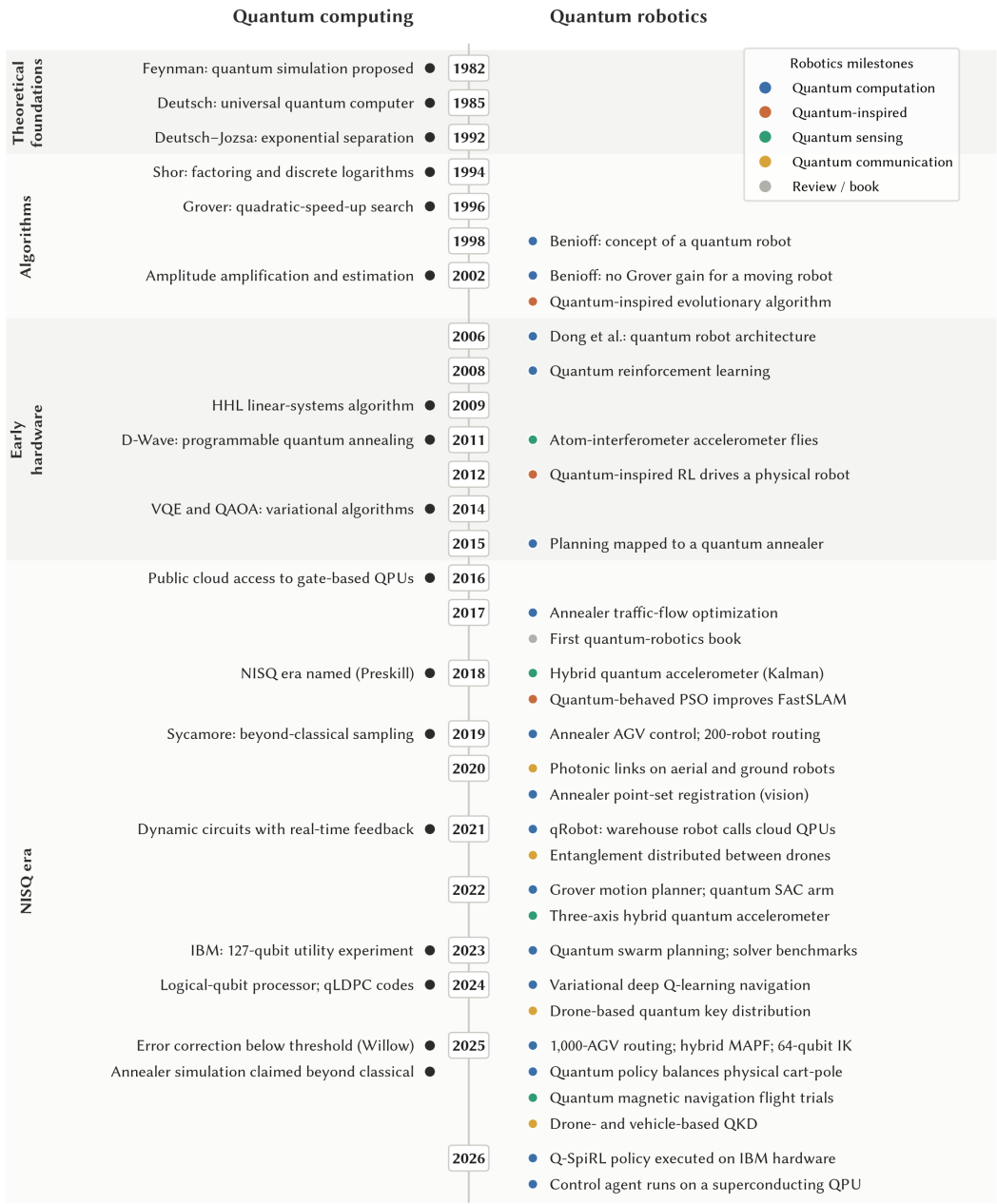


Fig. 1. Evolution of quantum computing and quantum robotics, 1982–2026, across the four eras used in this survey. Robotics milestones are colored by technology: quantum computation, quantum-inspired classical computation, quantum sensing, and quantum communication.

3.1 Research Questions

The survey addresses five research questions.

- **RQ1.** Which quantum computing methods have been applied to which robotic functions (optimization, planning, learning, sensing and communication, and decision-making)?
- **RQ2.** What level of evidence supports each application, from theoretical analysis to end-to-end deployment on physical robots?
- **RQ3.** Which reported benefits are genuinely quantum, and which are obtained by quantum-inspired classical algorithms or removed by dequantization?
- **RQ4.** Which hardware and deployment constraints (qubit counts, noise, latency, quantum-classical communication, and size, weight, power, and cooling) limit the use of quantum computing in robotics today?
- **RQ5.** Which directions offer realistic benefit in the near term (2026–2030), and what research is needed for the longer term?

3.2 Search Strategy

Searches were run in IEEE Xplore, the ACM Digital Library, Scopus, Web of Science, SpringerLink, ScienceDirect, and arXiv (categories quant-ph, cs.RO, and cs.LG), and we followed the references of key papers and earlier surveys backward and forward [113, 132, 160, 178]. The core search string combined a quantum block (for example, “quantum computing”, “quantum annealing”, QAOA, “quantum reinforcement learning”, “quantum-inspired”, “quantum sensing”, “quantum cognition”), a robotics block (for example, robot*, “autonomous vehicle*”, swarm, UAV, AGV, manipulator), and an application block (for example, planning, navigation, localization, “task allocation”, routing, control, learning, “decision making”, SLAM, kinematics). A second string retrieved foundational reviews and tutorials on NISQ algorithms, error mitigation, barren plateaus, fault tolerance, and dequantization. We soon found that a good part of the quantum-robotics work is published outside robotics venues, so we also searched the application areas where it tends to appear: drones and UAVs, multi-agent and autonomous mobility, autonomous driving, control benchmarks such as the cart-pole, and automated storage and retrieval. The search covered publications from 1982 onward. We closed the corpus in October 2026; the most recent study it contains was posted in August 2026. The exact strings for each database are listed in the supplementary material.

3.3 Eligibility Criteria

Studies were included if they (I1) proposed, evaluated, or reviewed a quantum or quantum-inspired method applied to a robotic function; (I2) provided foundational algorithms, hardware, or theory on which such methods depend; or (I3) supplied the classical robotics baselines needed for comparison. Studies were excluded if they (E1) used “quantum” only metaphorically, without a quantum-mechanical formalism or algorithm; (E2) were not available in English; (E3) were duplicates or earlier versions of an included study, in which case the most complete version was retained (for example, the journal version of Hohenfeld et al. [77] rather than its preprint); or (E4) addressed quantum communication or cryptography without a robotic application. Studies that satisfy I1 are the *primary studies*. Studies that satisfy only I2 or I3 are cited as *foundational works* and were not graded. Quantum computer-vision studies that solve perception problems on quantum hardware, but without a robotic application, fail I1; we discuss the six most relevant in Section 10.4 as related work and were not graded.

3.4 Study Selection

We ran the searches iteratively, refining the strings as the scope of the review settled, and screened candidates first on title and abstract and then in full text against the criteria above. We did not log how many records each search returned or how many were excluded at each stage. We therefore cannot give the record counts of a complete PRISMA 2020 flow diagram; the flow diagram in the

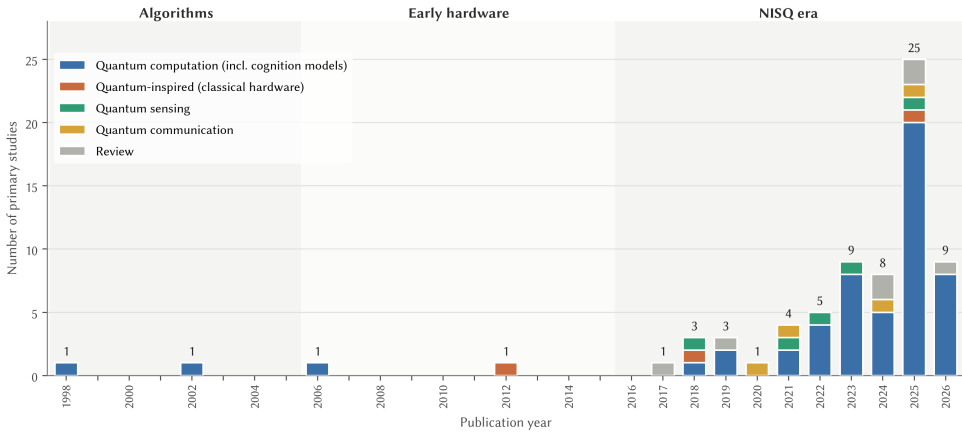


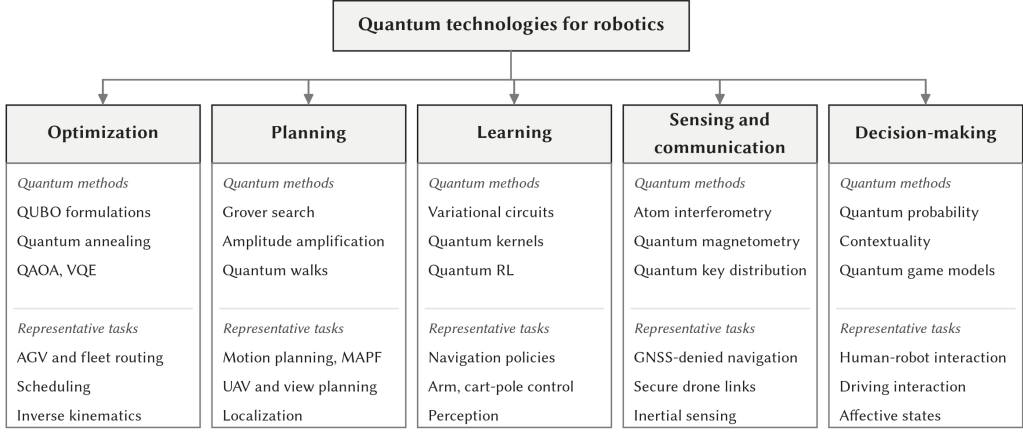
Fig. 2. Primary studies by publication year and technology type ($n = 72$, including the 7 reviews), from the first quantum-robot paper in 1998 to 2026, with the eras of Figure 1 shaded. The 2026 bar covers a partial year.

supplementary material shows the stages we followed and the counts we can document. The final corpus comprises 72 primary studies (65 empirical or theoretical studies and 7 reviews) and 107 foundational works, and six related quantum computer-vision studies are discussed separately (Section 10.4). Figure 2 shows how the field has evolved. Before 2006 the record holds only Benioff’s two theoretical papers. The useful quantum algorithms had only just appeared, and there was no programmable hardware to run them on, so robotics work could only be conceptual. Studies became regular once D-Wave annealers (2011) and cloud access to gate-based processors (2016) arrived. The growth since then is steep: 51 of the 72 primary studies appeared in 2023 or later, 25 of them in 2025 alone.

When we retrieved the full texts, we checked every PDF against its bibliographic record. A few files turned out to be mislabeled or to carry wrong metadata, and we corrected these before extraction. All bibliographic records were then verified against publisher, DOI, or arXiv records.

3.5 Data Extraction and Quality Assessment

For each primary study we extracted the bibliographic data, robotic task, quantum method, quantum type, hardware or simulator, experimental setup and baselines, key quantitative results, and stated and unstated limitations. Robotics application studies were additionally appraised against seven quality criteria (problem formulation, strength of the classical baseline, experimental realism, reporting of hardware and noise, performance metrics, reproducibility, and critical discussion of limitations), each judged as not met, partly met, or met. We chose not to add the judgments into a total score. A total hides which criterion a study fails, and the studies differ too much for totals to be compared. Instead we report the unmet criteria as the study’s principal limitation, in the evidence table and in the text of Sections 4 to 10, and they feed the assessment at the end of each section. The per-study judgments are given in the supplementary material. Two criteria fail most often: only 12 of the 59 appraised studies compared against a state-of-the-art, tuned classical baseline, 19 used no classical baseline at all, and only 6 were tested on a physical robot or in a realistic field setting. Reproducibility is the third weak point: of the 56 studies whose full texts we could check, only 14 released their code or data. The first author screened the records, extracted the data, and assigned all evidence levels, execution tags, and appraisal judgments; with a single reviewer there were no disagreements to resolve, and inter-rater agreement could not be measured (Section 3.7).



Applied to every study: quantum or quantum-inspired, NISQ or fault-tolerant, evidence level, where the quantum part ran, dominant deployment constraint

Fig. 3. Taxonomy of quantum technologies for robotics: robotic functions (top), the quantum methods applied to each (middle), and representative robotic tasks (bottom).

3.6 Classification Scheme

Each primary study was classified along a robotic-function taxonomy, an evidence hierarchy with a quantum-execution tag, and five further dimensions.

Robotic-function taxonomy. Each study was assigned to its primary robotic function: *optimization* (task allocation, routing, scheduling, and kinematics), *planning* (path and motion planning, view planning, and localization posed as search), *learning* (supervised quantum machine learning and reinforcement learning), *sensing and communication* (inertial and magnetic navigation sensors and quantum key distribution between robots), and *decision-making* (quantum-probability and cognitive models of choice). Figure 3 shows the resulting taxonomy, the quantum methods applied to each function, and representative robotic tasks; it also structures Sections 4 to 10.

Evidence hierarchy. Evidence was graded on six ordered levels, defined in Table 2. The hierarchy separates what has been *argued* (L1), *simulated* (L2), *executed on quantum hardware* (L3), *coupled to a robot or robot simulator in a closed loop* (L4), *demonstrated on a physical robot, vehicle, or mechanical control system* (L5), and *deployed end to end with a measured advantage over a classical baseline* (L6). A study's level is the highest level reached by its own experiments, not by the components it builds on.

Quantum-execution tag. The levels grade how far a study is integrated with a robot. They do not say, on their own, whether the quantum part ran on quantum hardware. We learned this the hard way while grading: L4 does not presuppose L3, because a study can couple a *simulated* quantum circuit to a simulated robot and reach L4 without ever using a quantum processor. Every study therefore also carries one of five execution tags: *none* (analysis or conceptual model only), *simulated* (quantum circuits, annealing, or quantum sensors simulated classically), *QPU* (at least part of the computation run on a gate-based processor or quantum annealer), *quantum sensor or link hardware*, and *classical* (quantum-inspired algorithms on conventional hardware). Level and tag are reported together (for example, L4/simulated), and Table 4 cross-tabulates them. We considered replacing the single scale with two independent axes, one for robot integration and one for quantum execution.

Table 2. Evidence hierarchy used to grade the primary studies, with the number of studies at each level (65 studies; 7 reviews excluded).

Level	Name	Definition	n	Examples
L1	Theoretical	Analysis, complexity argument, or conceptual model only	6	[13, 78, 99]
L2	Simulation	Quantum circuits or annealing simulated on classical computers	19	[30, 97, 145]
L3	QPU experiment	At least part of the method executed on real quantum hardware	23	[34, 100, 126]
L4	Robot-in-the-loop	Quantum computation coupled in a closed loop to a robot or high-fidelity robot simulator	10	[77, 136, 154]
L5	Physical demonstration	Behavior demonstrated on a physical robot, vehicle, or mechanical control system	6	[45, 101, 157]
L6	End-to-end deployment	Field operation in a realistic setting with a measured advantage over a classical baseline	1	[117]

We kept the ordered scale because the first question a roboticist asks of a result is how close it is to a working robot, and the levels answer that directly; the tag then answers the separate question of whether quantum hardware was involved. The cost of this choice is that the levels are not a ranking of quantum evidence. An L4/simulated study is closer to a robot than an L3/QPU study but says less about quantum hardware, and we compare studies on both dimensions throughout.

Further classification dimensions. Studies were further classified by (i) *quantum type*, genuinely quantum or quantum-inspired (Section 1.2); (ii) *method family*, such as QUBO formulations with annealing or QAOA, Grover search and amplitude amplification, variational circuits, quantum kernels, quantum-probability models, and quantum sensors; (iii) *hardware*, such as gate-based superconducting, trapped-ion, or photonic processors, quantum annealers, quantum-inspired digital annealers, or none; (iv) *regime*, NISQ-compatible or requiring fault tolerance; and (v) the *dominant deployment constraint*, such as qubit count, noise, latency, the input/output bottleneck, or size, weight, power, and cooling (SWaP-C).

3.7 Threats to Validity

Several threats apply to this survey. *Coverage*: the field moves quickly, and its terminology is loose (“quantum”, “quantum-inspired”, and “quantum-like” are used interchangeably). Several key results are still preprints [3, 117]. Broad search strings, seven databases, and reference chasing reduce these risks, but studies not in English and unpublished industrial work are missing. *Publication bias* probably favors positive results, so the actual rate of null or negative results is likely higher than reported here. *Classification*: evidence levels were assigned from the information reported, which is often incomplete on hardware, shots, and latency; when in doubt, we assigned the lower level. *Selection and grading*: one reviewer screened, extracted, and graded every study, and the number of records at each screening stage was not logged, so neither the completeness of the search nor the reliability of the grading can be measured directly. To make this checkable, we publish every evidence level, execution tag, and appraisal judgment in the supplementary material and dataset, so that readers can verify or contest each one. *Synthesis*: the heterogeneity of tasks, metrics, and baselines prevented a quantitative meta-analysis, so the synthesis is narrative and structured by the taxonomy. For a few records the analysis relied on the published abstract and supplementary material rather than the full text. *Recency*: 16 of the 72 primary studies were preprints when we

graded them. We graded them by the same rules as published work and flag them in the evidence table, but their results may change on publication. With 34 studies from 2025 or 2026 alone, the field moves faster than any survey can follow. The evidence levels describe the record at the time of our search, not the limits of the technologies.

4 QUANTUM OPTIMIZATION FOR ROBOT COORDINATION, SCHEDULING, AND KINEMATICS

4.1 Methods

Most robotic optimization problems addressed with quantum methods are first written as QUBO problems or, equivalently, Ising models (Section 2.1). Lucas [104] catalogued Ising formulations of Karp's 21 NP-complete problems and many others, including the traveling-salesman, coloring, and scheduling problems that underlie routing and task allocation. A QUBO can be solved by quantum annealing on superconducting flux-qubit hardware [81] or by gate-based variational algorithms. QAOA [52] alternates p layers of problem and mixing unitaries; its solution quality is non-decreasing in p and converges to the optimum as $p \rightarrow \infty$, but its single-layer guarantee on 3-regular MaxCut (0.6924) is weaker than that of the best classical algorithms. Hadfield et al. [71] generalized QAOA into the Quantum Alternating Operator Ansatz, whose mixers keep the state inside the feasible space and so enforce hard constraints without penalty terms. Zhou et al. [183] showed that optimal QAOA parameters follow transferable patterns that reduce the parameter search from exponential to polynomial effort in p , and Barkoutsos et al. [10] showed that optimizing the conditional value-at-risk of the samples, rather than their mean, improves convergence on combinatorial problems. Annealers were tried on robot-like planning early. Rieffel et al. [143] mapped navigation-type and scheduling-type planning problems to QUBOs on a D-Wave device and found that the problem mapping, the embedding, and the annealing schedule all had a strong effect on performance. Neukart et al. [118] used a hybrid annealer workflow to redistribute cars across a congested road network. Yarkoni et al. [179] review industrial uses of annealing more broadly.

Expectations need to be set carefully here. No exponential speed-up has been proven for NP-hard optimization, and QAOA and annealing are heuristics whose advantage, if there is one, has to be shown by experiment. From noisy simulations against a state-of-the-art MaxCut solver, Guerreschi and Matsuura [70] estimated that a runtime crossover would require several hundred to a few thousand qubits. The expected benefit for robotics is therefore faster good-enough solutions to large allocation and routing problems at scales beyond those of today's devices.

4.2 Multi-Robot Fleets and Automated Guided Vehicles

Quantum optimization is the largest and most hardware-tested cluster of robotics studies. Most of it concerns fleets of AGVs and transport robots, formulated as QUBOs and solved on D-Wave annealers or hybrid solvers.

Ohzeki et al. [123] formulated the collision-free control of AGVs in a factory as a QUBO in which binary variables select one route per vehicle from a set of candidates, and quadratic penalties forbid multiple selections and the simultaneous occupation of road segments or intersections; the QUBO is re-solved in every time period. Using a simulation of an actual DENSO plant with 10 AGVs, the authors compared a D-Wave 2000Q annealer, the Fujitsu Digital Annealer, and the Gurobi solver against the plant's rule-based control. The optimization-based control produced smoother, collision-free traffic. D-Wave returned plans quickly but handled only small instances (the largest QUBO had 60 variables) and needed post-processing to discard constraint-violating samples, whereas the digital annealer and the classical solver handled larger problems efficiently.

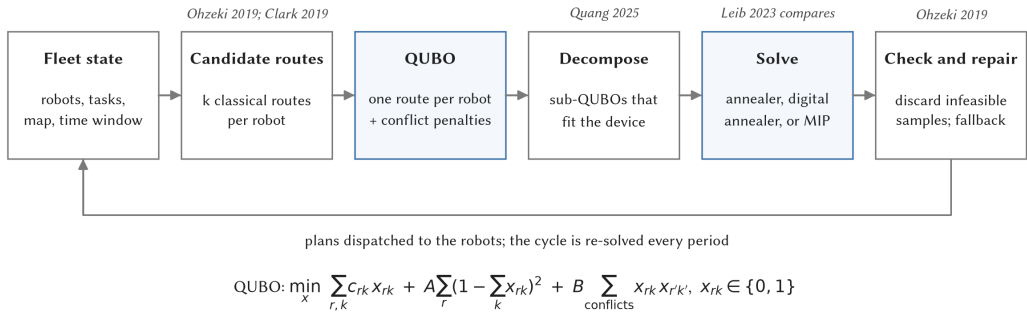


Fig. 4. The annealing pipeline shared by the fleet-routing studies. Each vehicle selects one route from a candidate set, penalty terms forbid double selection and conflicting road use, large instances are decomposed, and returned samples are checked and repaired before dispatch; the loop repeats in every planning period. Redrawn and generalized from the formulations of Ohzeki et al. [123], Clark et al. [34], and Quang et al. [136], with the solver comparison of Leib et al. [100].

The main lesson of this industrially grounded study is that the QUBO formulation, rather than the quantum hardware, delivered most of the benefit. The same pipeline, from fleet state to a repaired plan, recurs, with variations, across the annealer studies of fleet routing (Figure 4).

Quang et al. [136] scaled QUBO-based AGV routing to commercial warehouse size. They worked with the operating system of a commercial AGV product. A real-time generator proposes candidate routes, a priority-weighted cost function handles safety, and dynamic clustering splits the fleet-wide problem into sub-QUBOs small enough for current annealers; these are then solved on D-Wave hardware or on classical and quantum-inspired solvers. On a 154 m × 298 m layout with 8,242 tags, real warehouse request data, and an AGV simulator running up to 1,000 vehicles, route generation reduced the number of optimization variables by 96% for small and medium systems and by 78% for large ones. Clustering kept every subproblem below 10,000 variables, and a complexity-based formulation cut sampling time by about 28%. This is the largest robotics application of quantum annealing we found. Most of its gains, though, come from decomposition and problem engineering, and the paper does not show that the annealer beats classical solvers on the same subproblems.

Clark et al. [34] asked whether current annealers could route many robots on a grid in real time. A classical computer generates candidate paths for each robot, and a D-Wave 2000Q annealer solves the combinatorial problem of choosing one conflict-free path per robot, usually the most time-consuming step classically. Benchmarks with up to 200 robots showed that valid multi-robot routing solutions can be produced with existing annealing hardware. “Real time”, however, refers to the anneal; end-to-end latency, including cloud access and embedding, remains the bottleneck [139].

Three studies put quantum and classical solvers side by side. Śmierzchalski et al. [156] mapped AGV scheduling in a cramped factory onto a railway-style scheduling model and solved it with D-Wave’s hybrid solvers. Instances with up to 21 AGVs, where deadlocks are a real risk, were solved in seconds, with results comparable to the classical CPLEX solver. The authors themselves point out the catch: the hybrid solver is proprietary and mixes classical heuristics with occasional calls to the quantum processing unit (QPU), so the quantum contribution cannot be isolated. The same caveat applies to much of the hybrid-solver literature. Leib et al. [100] carried out one of the most rigorous comparisons in the corpus, on a transport-robot scheduling problem from a BASF high-throughput laboratory. With three different models of the problem, they compared D-Wave’s hybrid annealer, Fujitsu’s quantum-inspired Digital Annealer and its hybrid framework, and Gurobi

on both solution quality and end-to-end runtime. On smaller instances Gurobi reached comparable quality 10 to 100 times faster. On the largest instances the Fujitsu hybrid found good solutions within minutes where Gurobi hit its time limit, while D-Wave showed only limited promise. For robot scheduling today, the practical benefit of “quantum” optimization therefore comes mostly from quantum-inspired digital hardware. Finally, Atchade-Adelomou et al. [6] presented qRobot, a proof of concept in which the joint order-batching and picking-route problem of a warehouse robot is modeled as a capacitated vehicle-routing problem, translated into a QUBO, and solved with VQE and other hybrid algorithms. The robot’s onboard computer, a Raspberry Pi 4 with an ARM64 quantum software stack, dispatches jobs to cloud back ends including IBM Quantum, Amazon Braket, and D-Wave. The system returned valid routes for small instances. Its value is mainly the end-to-end architecture: the instances were very small, there was no systematic comparison with mature vehicle-routing solvers, and cloud queueing latency was not measured. Windmann [176] scheduled the transport jobs of automated storage and retrieval machines as an asymmetric QUBO solved with QAOA on an IBM Q System One processor, and measured how running time and qubit requirements grew with instance size. As in the annealer studies, only small instances fit on the device.

4.3 Industrial Trajectories, Assembly, and Inspection

Beyond fleets, the same approach has been applied to robot trajectory sequencing, assembly lines, and inspection paths. Schuetz et al. [148], from the Amazon Quantum Solutions Lab with BMW Group, sequenced robot tasks in automotive body and paint shops. Their pipeline rests on a classical biased random-key genetic algorithm, strengthened by model stacking, ensembles, and path relinking, and it consistently beat greedy baselines on industry-scale data. Quantum annealing on Amazon Braket was tested as a plug-in module. We found the paper a realistic template for a “quantum-ready” industrial pipeline, and it is honest that the performance comes from the classical algorithm.

Willmann et al. [175] formulated robotic assembly-line balancing as a QUBO and solved it on a D-Wave Advantage 4.1 processor. For a small case (four tasks, two workstations, two equipment types), the annealer found the optimal configurations among its samples, but the full workflow, including QPU communication, took about 10,110 s, compared with 0.03 s for an exact classical solver. Negative results like this one are rare and useful. It puts a number on how much the overheads outside the anneal (queueing, communication, repeated sampling) dominate the end-to-end runtime of a small industrial problem. Osaba et al. [124] applied D-Wave’s hybrid solvers to plan the trajectories of an inspection robot from CAD models, modeled as a three-dimensional, open-route traveling-salesman problem on an incomplete graph. Across five industrial cases, the nonlinear hybrid solver achieved quality competitive with Gurobi and Google OR-Tools with reduced computation time. It is one of the few studies to report a runtime benefit on a robotics task. Given the short-paper format, the proprietary solver, and the thin runtime detail, we treat it as preliminary.

4.4 Kinematics

Two recent studies execute robot kinematics on quantum hardware. Otani et al. [126] encoded the orientation of each link of a manipulator in a qubit through rotation gates, with entangling gates (R_{XX} , R_{YY} , R_{ZZ}) representing the coupling between parent and child links, so that the circuit reproduces the forward kinematics and a classical optimizer (COBYLA) searches for joint angles that reach a target pose. On a two-link manipulator with three rotational degrees of freedom per joint, entanglement reduced the positioning error and accelerated convergence relative to a circuit without entanglement, both in simulation and on the 64-qubit RIKEN/Fujitsu superconducting

computer. The problem is tiny (two qubits) and there is no comparison with standard inverse-kinematics (IK) solvers, so this is an early hardware demonstration, not evidence of advantage. Salloum et al. [146] instead reformulated planar IK as a QUBO by discretizing joint angles into one-hot binary variables, and solved it on D-Wave annealers with the Pegasus and Zephyr topologies. They defined time-to-solution rigorously and compared embedding strategies: global embedding on Zephyr used the fewest physical qubits, and the hybrid solver reached lower time-to-solution than classical QUBO baselines on larger instances. The authors state explicitly that the method is not superior to state-of-the-art continuous IK solvers. Read together, the two studies show that kinematics can run on today's hardware, but only in discretized or toy form, where classical continuous solvers are still stronger.

4.5 Assessment

Classical baselines are decisive for judging these results. Sampling-based planners such as RRT [98], RRT* and PRM* [86], and integrated task-and-motion planners [60] solve continuous problems that current QUBO formulations represent only after coarse discretization. Where studies did compare against mature commercial solvers, the picture is consistent. Gurobi or CPLEX matched or beat the annealer on small and medium instances [100, 123, 156], an exact classical solver was orders of magnitude faster on a small assembly problem [175], and advantages on the largest instances came from quantum-inspired digital annealers rather than quantum hardware [100].

Where the evidence stands. Evidence reaches L3 and, for fleet routing, L4 in simulation. Annealers have produced valid collision-free plans for up to 200 robots [34], sub-QUBOs of a 1,000-AGV fleet have been routed through a commercial AGV operating system in simulation [136], and kinematics has been solved on annealers [146] and on a 64-qubit gate-based processor [126]. No study reports a robust runtime or quality advantage over the best classical solver. One short study reports reduced computation time for inspection paths [124], and it still awaits confirmation.

What holds the area back is problem size and embedding. Penalty-based QUBOs inflate qubit counts, annealer connectivity forces embedding overheads, and classical and quantum-inspired digital solvers [68] currently solve the same QUBOs at much larger scale. Most of the reported benefit comes from formulating the problem well, not from the quantum hardware.

5 QUANTUM SEARCH FOR PLANNING, NAVIGATION, AND LOCALIZATION

5.1 Methods

Grover's algorithm finds a marked item among N using $O(\sqrt{N})$ oracle calls [69], and amplitude amplification generalizes this to any procedure that succeeds with probability a , needing $O(1/\sqrt{a})$ repetitions instead of $O(1/a)$ [20]. Amplitude estimation yields a near-quadratic speed-up for the Monte Carlo estimation of expected values [115], which is directly relevant to particle-filter localization [40] and probabilistic planning. Quantum walks provide quadratic speed-ups for search and for the deliberation of learning agents [128], and coherent distance estimation gives polynomial query reductions for nearest-neighbor classification and clustering [174], a primitive used in sampling-based planners and place recognition. These speed-ups count oracle queries, not wall-clock time, and they need fault-tolerant hardware. Benioff [14] had already shown that a quantum robot gains nothing from Grover search when it has to move through the two-dimensional region it searches, and that warning still applies to the planning studies below.

5.2 Motion Planning and Swarm Planning

Chella et al. [30] formulated robot motion planning as quantum search. The robot's world, admissible actions, and goal conditions are expressed as a production system, and the tree of possible action

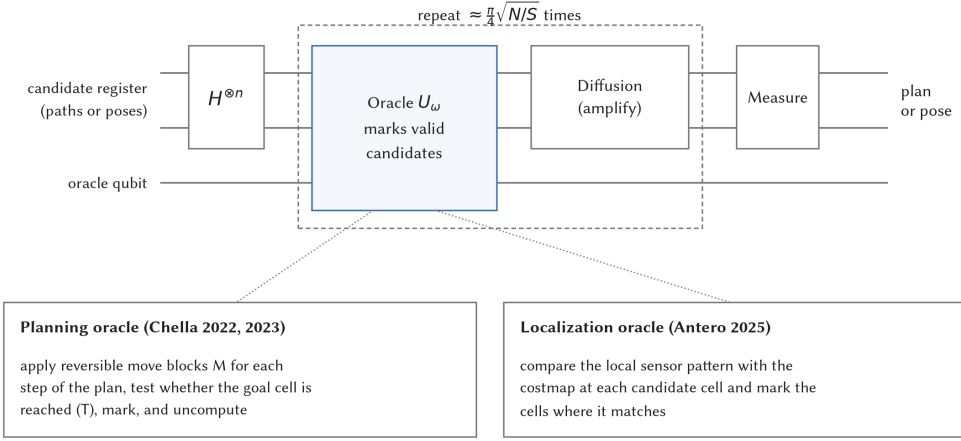


Fig. 5. Grover search as used for robot planning and localization. After a uniform superposition, the oracle marks goal-reaching action sequences [29, 30] or map cells whose stored pattern matches the current sensor view [4], and the diffusion step amplifies the marked states; about $\frac{\pi}{4} \sqrt{N/S}$ repetitions are needed for N candidates and S solutions. Redrawn from the circuit descriptions in those studies.

sequences is encoded in a reversible Boolean oracle so that Grover's algorithm can find a goal-reaching sequence with $O(\sqrt{N})$ oracle calls instead of exhaustive $O(N)$ search. The authors detail the translation of classical logic into reversible gates with ancilla qubits, analyze the success probability for single and multiple solutions, and introduce a qubit-reuse technique. All experiments ran on the noiseless Qiskit Aer simulator, which limited problems to about 27 qubits. As a worked mapping of planning onto a gate-based algorithm, the paper is instructive. Its limits are just as clear. Qubit requirements grow with plan depth, the oracle assumes a fully known map, and there is no comparison with heuristic planners such as A^* or with sampling-based methods. Chella et al. [29] extended the planner to swarms of simple robots in a search-and-rescue scenario modeled on ant foraging. Each robot's position and perceived proximity to the target are encoded in qubits, pairwise information exchange is a quantum circuit that decides which peer to follow, and each robot plans its next moves with a Grover circuit. In Qiskit simulations the quantum swarm reached the target faster than two classical ant-behavior simulations. The swarms were small, however, the circuits noiseless, and the baselines generic ant-colony simulations, not state-of-the-art multi-robot planners. Lathrop et al. [97] started from the sampling-based planners themselves. They recast sampling-based motion planning as a database-oracle problem and solved it with Grover-type quantum search, with a quantum full path search for sparse environments and a quantum RRT for dense ones. In a classical simulation of the quantum algorithm, q-RRT showed the expected quadratic speed-up over classical RRT on two-dimensional random lattices. Of all the studies we read, this one comes closest to the planners robots actually use. Even so, the speed-up was measured in a classical simulation of the quantum algorithm, and only fault-tolerant hardware would turn it into a real one. Figure 5 shows the common structure of these planners and of the Grover localizer discussed below: only the oracle changes between applications.

Nigatu et al. [122] combined quantum machine learning with Grover search for manipulator optimization: a parameterized circuit is trained to approximate the forward kinematics and then used to build the oracle with which Grover's algorithm searches the configuration space. In simulated

one-degree-of-freedom, two-degree-of-freedom, and dual-arm tasks, the method reported speed-ups of up to 93 times over classical optimizers such as Nelder-Mead as dimensionality increased. The speed-ups were measured in simulation and rest on Grover's quadratic query advantage, so the same fault-tolerance caveat applies. Sarkar et al. [147] proposed a hybrid quantum ant-colony optimization architecture for "quantum-enabled" mobile robots that must visit a set of targets, which converts classical data to circuits automatically and runs many ants in parallel. On TSPLIB instances and 50 random 16-node graphs in the Qiskit simulator, it produced optimal or near-optimal tours with an average error of about 7% relative to OR-Tools optima; accuracy degraded as the number of qubits grew, and runtime benefits are only projected.

5.3 Multi-Agent Pathfinding and UAV Navigation

Gerlach et al. [62] presented the first optimal hybrid quantum-classical algorithms for multi-agent pathfinding (MAPF), the core coordination problem of robot fleets. Building on branch-and-cut-and-price, the method decomposes conflicts between agents into conflict graphs and solves the resulting small, independent QUBO subproblems with a quantum or quantum-inspired solver. On standard MAPF benchmarks, including runs on real quantum hardware, the approach dominated earlier QUBO formulations and compared favorably with state-of-the-art classical MAPF solvers. Methodologically, this is the strongest multi-robot result in the corpus. The quantum component sits inside a provably optimal classical framework, and the baselines are strong. González Villasmil [66] formulated simultaneous multi-robot path planning as a single QUBO, using breadth-first-search preprocessing to remove more than 95% of the variables and time-windowed decomposition to keep problems tractable. On grids of up to 10×10 cells with four robots, solved by simulated annealing and simulated QAOA, it found near-optimal paths (a 5.3% path-length penalty in dense scenarios), but classical planners remained faster at all tested sizes. Rao et al. [138] designed a constraint-preserving QAOA formulation [71] for multi-vehicle coverage path planning, proved its validity, gave efficient constructions of the problem and mixer operators, and estimated the resources needed on current hardware, reporting improvements over depth-first-search exploration, which is a weak baseline.

Innan et al. [79] proposed QUAV, one of the first applications of QAOA to drone trajectory planning. The flight graph, obstacles, and no-fly zones are encoded in a cost Hamiltonian, geospatial coordinates are transformed so that paths are geographically accurate, and circuit depth scales linearly with the number of graph edges. In simulation and on IBM's ibm_kyiv processor, QUAV produced feasible trajectories despite noise and limited connectivity. The graphs are small, though, and there is no benchmark against classical planners. Osaba et al. [125] split real-world drone routing, with asymmetric costs, forbidden paths, and charging points, into a QAOA clustering phase and a routing phase solved on D-Wave annealers, and obtained feasible routes for three use cases of increasing complexity. Much of the work is done by the classical decomposition, and the quantum contribution is not isolated. Bang et al. [9] used a fine-tuned language model to turn natural-language mission instructions into QUBO constraints for multi-drone task assignment, and kept the assignment constraints inside the feasible space with W-state initialization and XY mixers; with CVaR-QAOA sampling, the decomposed problems recovered the global optimum in 96.3% of cases, although only in statevector simulation.

5.4 Perception-Driven Planning and Localization

Yu and Chen [180] proposed HQC-NBV, a hybrid quantum-classical approach to next-best-view planning, the choice of camera viewpoints for exploring or reconstructing an unknown scene. Viewpoint selection is encoded in a multi-term Hamiltonian, and a variational ansatz with bidirectional

alternating entanglement captures dependencies between viewpoint parameters. In simulated environments the method explored 7.9–49.2% more efficiently than classical approaches, and ablations attributed part of the gain to the entanglement structure. The circuits were simulated, though, and the baselines were not state-of-the-art view planners. Antero et al. [4] used Grover search for mobile-robot localization on a known two-dimensional map. The environment is encoded as a costmap of obstacle proximities, the robot’s local sensor view forms a perception pattern, and localization becomes the problem of finding where that pattern occurs in the map, which a Grover oracle solves with $O(\sqrt{N})$ queries instead of the $O(N)$ cell evaluations of grid-based Monte Carlo localization. The approach localized the robot correctly in simulation and was also run on cloud quantum hardware, but accuracy on real devices degraded as the map grew, and classical simulation becomes infeasible beyond about 40 qubits. The reported speed-up is therefore an asymptotic query count that presumes a large fault-tolerant device. It is a clear case of the distance between a theoretical advantage and a realized one.

Evidence level. Grover-based planning remains at L2 (noiseless simulation) [30, 122]. QAOA-based UAV planning [79] and Grover localization [4] reach L3 on very small problems, with accuracy degrading on real devices, and hybrid MAPF reaches L3 with strong classical baselines [62]. Apart from the last, no study compares against heuristic planners such as A^* , RRT, or RRT* [86], or against Monte Carlo localization on realistic maps [40].

In practice the obstacle is the oracle. Grover-type speed-ups are at most quadratic and apply to query counts, not wall-clock time, and building the reversible oracle for a map or plan tree requires many ancilla qubits and deep circuits, which exceed NISQ coherence budgets, and the quadratic gain is quickly consumed by error-correction overheads [58].

6 QUANTUM LEARNING FOR ROBOTS

6.1 Supervised Quantum Machine Learning

Quantum machine learning (QML) has developed along two lines [16, 150, 172]. The first relies on fault-tolerant linear algebra: quantum clustering [102] and the quantum support vector machine [140] promised exponential speed-ups given quantum random-access memory (qRAM). The second uses NISQ-compatible parameterized circuits: quantum circuit learning [114] and quantum neural networks for classification [53] train low-depth circuits with a classical optimizer, while the kernel view [149] interprets data encoding as a feature map into Hilbert space. Havlíček et al. [75] implemented a variational classifier and a quantum kernel estimator on IBM hardware, reaching 100% accuracy on synthetic data designed to suit the chosen feature map, and quantum convolutional neural networks [76] inserted random quantum filters into convolutional networks without showing an advantage over random classical nonlinear features.

We found surprisingly few direct applications of supervised QML to robot perception. (Annealing-based approaches to registration and tracking are discussed in Section 10.4.) The clearest perception study is Roh et al. [145], who designed a fast quantum convolution that loads several input channels with few qubits and used it for object detection on the KITTI driving dataset; the circuits are simulated, but the data are real. Petschnigg et al. [132] and Yan et al. [178] identify quantum image processing, quantum principal component analysis, and quantum support vector machines for object recognition and sensor fusion as candidate applications, but note that existing quantum image representations handle two-dimensional images, whereas robots mostly see the world as three-dimensional point clouds. The first QML study on real industrial robot data targeted robot software engineering: with PAL Robotics, Wang et al. [169] built a test oracle that predicts the expected behavior of an autonomous mobile robot for regression testing after software updates. Quantum reservoir computing alone failed to converge, but combining it with a small classical

network produced an oracle whose prediction error was about 15% lower than that of a classical neural-network baseline. The study points to offline software assurance as a realistic, latency-tolerant niche for QML in robotics. Earlier, Gonçalves [65] ran small quantum neural networks for a quantum robot interacting with a quantum target on IBM's five-qubit processor, in the tradition of robots acting on quantum environments [13].

Barren plateaus. McClean et al. [110] showed that the gradients of randomly initialized parameterized circuits vanish exponentially with the number of qubits. Cerezo et al. [27] showed that global cost functions cause barren plateaus even at shallow depth, whereas local costs keep gradients polynomial at logarithmic depth. Larocca et al. [96] unified the sources of barren plateaus (expressivity, global costs, entangled data, and noise) and raised the concern that architectures provably free of barren plateaus may also be efficiently classically simulable, which would remove their quantum advantage. Robot learning, with its high-dimensional inputs and often global rewards, runs straight into this limit.

Dequantization. Several claimed exponential QML speed-ups have been matched by classical sampling-based algorithms under the same input assumptions [33, 161], and the fine print of quantum linear-algebra speed-ups (efficient state preparation and limited read-out) rarely holds for robot data [1].

6.2 Quantum Reinforcement Learning

Quantum reinforcement learning (QRL) has three strands [49, 113]. The first is quantum-inspired: Dong et al. [46] represented actions as superpositions whose amplitudes are updated by Grover-like amplification, with action selection by measurement collapse. The second is agent-centered: projective simulation [22] models deliberation as a random walk over episodic memory, which a quantum walk can speed up quadratically [128]. The third strand, now the dominant one, replaces the neural networks of deep RL [158] with variational quantum circuits (VQCs). Its building blocks include VQC-based deep Q-learning [32], data re-uploading and trainable output scaling for Q-learning [155], quantum policy gradients with a proven learning separation on artificial tasks [80], gradient-free evolutionary training with tensor-network compression for high-dimensional inputs [31], and multi-agent QRL with centralized training and decentralized execution [181]. How the circuit is designed, from angle embedding to output post-processing, largely decides whether such agents learn continuous control at all [94].

Quantum-inspired navigation. Dong et al. [45] applied quantum-inspired RL (QiRL) to mobile-robot navigation. Each action is an "eigenaction" in a superposition whose amplitudes are learned; action selection mimics measurement collapse, so exploration and exploitation are balanced by the amplitudes, and a Grover-inspired rule amplifies rewarded actions. QiRL learned obstacle avoidance and goal reaching faster than temporal-difference learning in simulation and on a real two-wheeled robot, and was more robust to the choice of learning rate. The algorithm runs entirely on classical hardware, and its only baseline was plain temporal-difference learning. Still, it is one of the very few quantum-motivated learning methods ever tested on a physical robot.

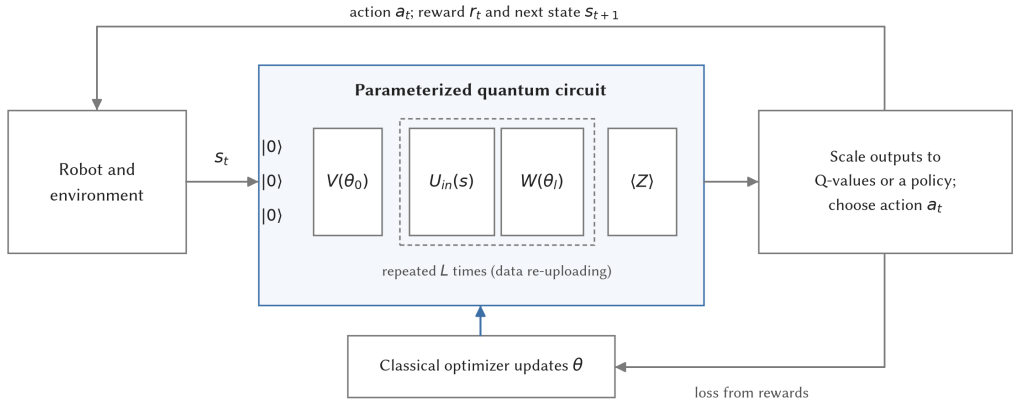
Variational QRL for navigation and control. Hohenfeld et al. [77] replaced the Q-network of Double Deep Q-Networks with parameterized circuits and trained a simulated Turtlebot 2 to navigate grid environments of increasing size and a larger dynamic environment with moving obstacles and lidar sensing. The circuits learned optimal policies with far fewer trainable parameters and beat classical networks of the same size. The full-size classical baseline, however, with at least ten times as many parameters, trained faster and more stably and learned better policies in the dynamic environment. Drăgan et al. [48] extended this line to continuous states and actions with VQC actors

and critics in Deep Deterministic Policy Gradient (DDPG). A simulated Turtlebot 2 controlled its wheel velocities from three lidar readings, orientation, and speed in warehouse-like mazes. The best quantum model (8 qubits, 5 layers, 851 parameters) solved the 3×3 and 4×4 mazes in all 30 trials but the 5×5 maze in only 10 of 30, whereas a classical DDPG with 136,451 parameters solved all three consistently. Sinha et al. [154] proposed Nav-Q for collision-free navigation of self-driving cars, in which only the critic of an actor-critic architecture is a hybrid quantum-classical network, so the trained policy runs classically at deployment, together with a qubit-independent data-encoding scheme for high-dimensional inputs. On the CARLA-CTS1.0 benchmark, Nav-Q trained more stably than its classical counterpart and, in some settings, converged faster with substantially fewer parameters; noisy simulation showed that hardware noise degrades training. Acuto et al. [2] studied a variational quantum Soft Actor-Critic (SAC) on a simulated robotic arm. Replacing only the actor gave the same learning curve as the classical SAC, but a fully quantum SAC with about 2,700 parameters converged where a classical SAC with about 3,000 parameters failed, and the classical agent needed about 274,000 parameters, roughly 100 times more, to reach comparable returns. Lokossou et al. [103] extended variational SAC to legged locomotion on the MuJoCo Walker2d environment with 17-qubit circuits in simulation, reaching an average return about 8% higher than a classical SAC with 41 rather than 1,250 learnable parameters; qubit limitations prevented a move to the full Humanoid environment.

Multi-agent QRL for drones and mobility. Park et al. [129] gave a fleet of UAVs that provide mobile network access a quantum centralized critic, which kept the number of qubits manageable as the fleet grew and trained faster than classical multi-agent baselines. Park et al. [130] applied the same actor-critic idea to cooperative autonomous mobility, using a projection-value measure to shrink the action space logarithmically. Both use simulated circuits and simulated environments.

Real-time control. Two recent studies take quantum policies out of simulation. Sun et al. [157] trained a variational policy offline and deployed it on a physical cart-pole driven by a Raspberry Pi and an Arduino. The policy balanced the pole, but only because the circuit was simulated locally: on a cloud QPU, each decision took seconds, far past the control deadline. Ngo et al. [119] went the other way and kept the QPU. Their single-qubit actor-critic agent ran on the VTT Q5 superconducting processor, and by programming the readout electronics directly instead of going through the standard software stack, they made circuit execution more than ten times faster, reaching up to 6.2 inference steps per second. That is still short of the tens of hertz a closed control loop needs. Between them, these two papers give the first real measurements of what it would take to put a QPU inside a control loop. The variational policies in this section share one template (Figure 6); they differ mainly in where the circuit sits and whether a QPU is ever queried.

Quantum-neuromorphic hybrids. Altrabulsi et al. [3] proposed Q-SpiRL, which combines spiking neural networks with an 8-qubit variational circuit for obstacle-aware navigation. Compared with tabular Q-learning, a classical multilayer perceptron, a classical spiking network, and a quantum-enhanced perceptron on grid worlds of up to 40×40 cells with static and dynamic obstacles, the quantum-enhanced spiking network gave the best overall trade-off, with up to 99% success, and the trained policy was run on IBM hardware as a feasibility check. The environments are discrete, however, and the policies are converted into Q-tables before evaluation. Taghavi and Farnoosh [159] combined VQCs for high-level policy exploration with spiking networks for low-latency motor control in a multi-agent RL framework; in a simulated forest with ten UAV agents it caused fewer safety violations than classical multi-agent baselines, although the share of the quantum component is not isolated.



Where the circuit sits in the studies

- the circuit is the Q-network
- only the critic is quantum; the deployed policy runs classically
- circuits form both actor and critic
- a simulated circuit controls a physical cart-pole
- the circuit runs on a quantum processor

Hohenfeld 2024, simulated Turtlebot

Sinha 2023, Nav-Q

Acuto 2022; Drăgan 2025

Sun 2025

Ngo 2026; Altrabulsi 2026

Fig. 6. The variational policy template common to the quantum reinforcement-learning studies: the robot state is encoded by data re-uploading, trainable layers are repeated L times, measured expectation values are scaled into actions or values, and a classical optimizer updates the parameters. Adapted from the architectures of Hohenfeld et al. [77], Sinha et al. [154], Acuto et al. [2], Drăgan et al. [48], Sun et al. [157], Ngo et al. [119], and Altrabulsi et al. [3]; the list on the right notes where the circuit sits in each study.

6.3 Assessment

Looking across the robotics studies, QRL has one consistent advantage, and it is *parameter efficiency*. Quantum policies match classical ones of equal size and reach comparable returns with one to two orders of magnitude fewer trainable parameters [2, 48, 77, 103], and Nav-Q reports more stable training [154]. Adequately sized classical networks, however, train faster and more stably and perform better in the hardest environments [48, 77], and a formal learning advantage exists only for artificial tasks [80]. But a smaller parameter count is not the same thing as shorter training, better sample efficiency, or lower energy use on real quantum devices.

Maturity. Supervised QML for robotics is still at L1–L2. Its hardware demonstrations use two to a few qubits and synthetic or down-sampled data [53, 75], and the robot software-testing oracle is the only study on real industrial robot data [169]. Variational QRL reaches L4, but only in simulated robot environments with idealized, noise-free circuits [2, 48, 77, 103, 154]. Q-SpiRL reaches L3 by running a trained policy on IBM hardware [3], and a control agent runs on a QPU at L3 [119]. At L5 there are two studies: the quantum-inspired QiRL on a physical robot [45], and a simulated variational policy balancing a physical cart-pole [157]. The latter is the only genuinely quantum computational method to have reached a physical system so far, and even there the circuit runs on a classical computer.

Why QRL has not reached real robots. What really limits robot RL is the cost of real-world samples [93], not the size of the network, and no QRL study shows better sample efficiency on a robot task. Training on real QPUs would also mean millions of circuit evaluations, each facing

the queueing latencies discussed in Section 8. Exponential QML speed-ups rely on qRAM state preparation that does not yet exist, and NISQ models face barren plateaus and encoding bottlenecks for high-dimensional sensor data.

7 QUANTUM SENSING AND COMMUNICATION FOR MOBILE ROBOTS

7.1 Quantum Navigation Sensors

Quantum sensing uses quantum systems, such as cold atoms, nitrogen-vacancy (NV) centers, and trapped ions, to measure physical quantities with sensitivity and stability beyond those of classical sensors [39]. Cold-atom interferometers provide accelerometers and gyroscopes with extremely low bias drift, and optically pumped magnetometers measure weak magnetic fields. Bongs et al. [19] reviewed the transition of atom-interferometric sensors from the laboratory to the field and identified SWaP-C, dynamic range, and robustness as the main engineering barriers.

Cheiney et al. [28] addressed the main obstacles to using cold-atom interferometers for navigation, namely their low sample rate and limited dynamic range. In their hybrid accelerometer, a matter-wave interferometer, which is extremely stable but slow, continuously estimates the bias drift of a fast classical accelerometer through a Kalman filter. Under experimentally simulated vehicle-like conditions, the hybrid sensor combined a 400 Hz bandwidth with a stability of 10 ng after 11 hours of integration. Wang et al. [170] approached the same problem with maximum-likelihood fusion. A co-located classical accelerometer resolves the phase wrapping of the quantum sensor, and the recovered quantum measurement in turn recalibrates the bias of the classical sensor. In a simulated one-dimensional inertial-navigation scenario, this reduced the navigation error. Wang et al. [171] assessed NV-center (quantum diamond) magnetometers as aids to inertial navigation where global navigation satellite systems (GNSS) are unavailable. They combined simulated magnetic measurements with probabilistic data association and Viterbi map-matching to estimate the navigation errors that current and near-term sensitivities would allow. These solid-state sensors could eventually be small enough for mobile robots. Atom interferometers have also left the laboratory. Geiger et al. [61] operated a matter-wave accelerometer on board an aircraft, and Templier et al. [164] built the first hybrid three-axis quantum accelerometer, which tracks the full acceleration vector at 1 kHz with a 50-fold stability gain over its classical triad and no longer needs to be aligned with a predefined axis.

Muradoglu et al. [117] demonstrated “quantum-assured” magnetic-anomaly navigation, in which quantum magnetometers measure the Earth’s crustal magnetic field and a denoising and map-matching suite compares the measurements with anomaly maps to provide passive, jamming-resistant position fixes without GNSS. In flight trials on a crewed aircraft (a Cessna 208B) from ground level to about 19,000 ft, the system achieved up to about 46 times lower positioning error than a velocity-aided strategic-grade inertial navigation system (INS), with a best final error of 22 m (0.006% of the distance flown) and at least an 11-fold advantage in every airborne trial. On a ground vehicle with public maps, the error was about 7 times lower than that of the INS. The system learns vehicle-specific parameters online and is compact enough for fixed-wing drones. This is the strongest field evidence we found that quantum technology already improves a robotic capability. Two qualifications apply: the benefit comes from quantum sensing, not quantum computing, and the result is a company-led preprint that still needs independent replication.

7.2 Quantum Communication between Robots

Khoshnoud et al. [87], with NASA JPL, implemented single-photon generation by spontaneous parametric down-conversion, polarization encoding, and single-photon detection on networks of aerial and ground robots, as a step toward quantum-secured and entanglement-assisted multi-robot

coordination. Tian et al. [167] reported drone-based quantum key distribution (QKD): a compact, polarization-encoded decoy-state BB84 transmitter on an octocopter distributed secret keys in real time over 200 m at an average rate of 8.48 kHz. Liu et al. [101] distributed entangled photons over 1 km, with one drone carrying the entangled-photon source and a second drone acting as an optical relay, and still violated the CHSH inequality ($S = 2.59 \pm 0.11$). Conrad et al. [35] built a modular, reduced-SWAP QKD transmitter and receiver and demonstrated drone-to-drone, drone-to-vehicle, and vehicle-to-vehicle quantum communication with finite-key secure rates of 1.6–20 kbps, supported by device models of the non-idealities of mobile operation. Quantum-secured links between moving robots have thus reached field demonstration. That puts them on a par with quantum navigation sensing and well ahead of quantum computation for robots.

Of all the branches covered here, this one is the most mature. On the strength of one company-led preprint, magnetic-anomaly navigation reaches L6, with end-to-end field operation on aircraft and a ground vehicle and a measured advantage over a strategic-grade INS [117], and QKD and entanglement distribution between drones and vehicles reach L5 [35, 101, 167].

What limits it now is mostly engineering: size, weight, power, and cost, together with dynamic range. Cold-atom sensors are still bulky, power-hungry, and limited in bandwidth, and they must be fused with classical sensors by Kalman-type estimators [28, 170]. Shrinking them for small robots is work in progress [19]. QKD links are limited by range, alignment, and weather.

From aircraft to robots. None of the field results above involved an autonomous robot closing its own control loop on the quantum measurement, and that is the step that remains. For navigation, the integration path is clear in principle. A magnetic-anomaly fix is a periodic absolute position measurement, so it enters a robot's estimator in the same way as a GNSS fix, as a measurement update in a Kalman filter or a factor in a pose graph, while a cold-atom accelerometer enters as a low-drift inertial input whose bias is estimated alongside the classical sensor [28, 170]. What is missing is evidence on the platforms where GNSS-free navigation matters most to roboticists. Fixed-wing drones and ground vehicles are within reach of the demonstrated sensor [117]; small multirotors are harder, because their motors and power electronics sit close to the sensor and add the kind of platform interference that the denoising must remove. The method also depends on the quality of the anomaly map, which is coarse in many regions and of little use indoors. For communication, the open question is what a quantum-secured link buys a robot team. Key rates in the kilobit-per-second range [35, 167] are enough to refresh symmetric keys for command authentication, but no study has yet used such keys inside a multi-robot coordination protocol or measured what happens to the team when the link drops.

8 HYBRID QUANTUM-CLASSICAL ARCHITECTURES AND DEPLOYMENT CONSTRAINTS

8.1 Where the Quantum Processor Sits in the Robot

Every practical quantum-robotics system reviewed here is hybrid. Variational algorithms pair a QPU that prepares and measures parameterized states with a classical optimizer [15, 26, 111], and error mitigation must be co-designed with the algorithm [50]. On a robot there is one more decision to make: the classical controller has to choose which subproblems to offload, when, and to which processor. In the studies we reviewed, the QPU sits in one of three places (Figure 7): (i) *offline*, as a training-time accelerator whose output is a classical policy, as with Nav-Q's quantum critic [154]; (ii) in a *slow outer planning loop*, solving periodic allocation or routing problems, as in AGV control re-solved in every period [123], multi-robot routing [34], and warehouse picking [6]; and (iii) *inside the decision loop*, as a policy or search oracle queried at run time, as in Q-SpiRL [3] and Grover-based localization [4]. Only positions (i) and (ii) are compatible with current latencies.

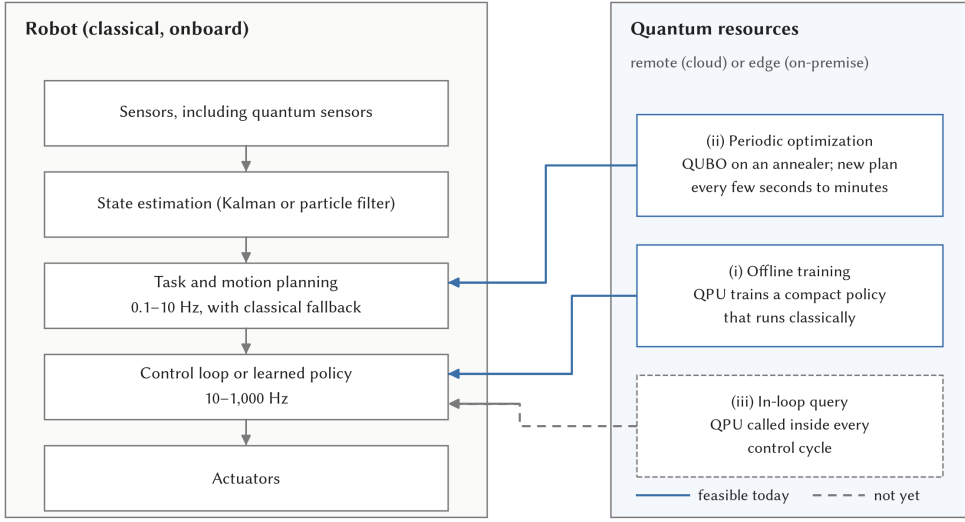


Fig. 7. Positions of quantum resources relative to a robot’s classical control stack. Offline training (i) and periodic, asynchronous optimization with a classical fallback (ii) are feasible today; an in-loop quantum query (iii) is not.

Deployment models. Current superconducting processors require dilution refrigerators [92], so robots access QPUs through the cloud (IBM Quantum, Amazon Braket, and D-Wave Leap), as in qRobot’s Raspberry-Pi-based controller [6]. In the medium term, the realistic model is an *edge* deployment, an on-premise QPU serving a factory or warehouse fleet, in line with the wider move to couple QPUs tightly with high-performance computing centers [44]. An onboard QPU on a mobile robot, as Benioff [13] and Dong et al. [47] imagined it, is out of reach with present technology [108].

Execution modes. Batch execution, such as offline training or overnight planning, tolerates any latency. Asynchronous execution, in which a QPU refines a plan while a classical fallback keeps the robot running, tolerates seconds to minutes. Real-time execution, with a QPU call inside a millisecond control cycle, is not possible today.

Architecture patterns. Three patterns recur: classical candidate generation followed by quantum selection, in which candidate paths are generated classically and their combination is chosen by annealing [34, 123]; quantum function approximation within classical RL loops [48, 77]; and quantum sensing fused with classical estimation [28]. Newer capabilities extend these patterns. Dynamic circuits with mid-circuit measurement and real-time classical feed-forward [36] bring classical decision logic inside a quantum computation, which closed-loop use will require. Circuit cutting [162] spreads large circuits across several small QPUs and reconstructs the result classically. Error-mitigated variational pipelines [50] and CVaR objectives [10] improve solution quality per shot.

8.2 Hardware Constraints

Platforms and qubit counts. Superconducting processors lead in qubit count and gate speed but require millikelvin cryogenics [92]; trapped-ion devices offer higher fidelities and all-to-all connectivity at slower gate speeds, as used by Widdows et al. [173]; and photonic and annealing

platforms serve specialized workloads [81, 131]. It is striking how small the robotics circuits are. Gate-based experiments used one to eight qubits, even on processors with up to 64 [3, 79, 126], whereas annealer studies solved QUBOs for up to 200 robots directly [34] and for 1,000-AGV fleets only after decomposition into subproblems below 10,000 variables [136].

Noise, mitigation, and fault tolerance. Gate errors and decoherence limit circuit depth [135]. Error mitigation, including zero-noise extrapolation and probabilistic error cancellation [163], improves expectation values on real processors [85], but its sampling overhead grows exponentially with circuit size [25]. In robotics, noise degraded localization accuracy on real devices [4] and training in noisy simulation [154]. Algorithms with provable speed-ups need logical qubits protected by error correction [165]. At realistic error rates, surface codes need on the order of a thousand physical qubits per logical qubit [58]. Low-density parity-check codes cut this by about an order of magnitude [21], and below-threshold operation has now been demonstrated [67]. Even so, resource estimates for industrially useful chemistry already call for large fault-tolerant machines running for a long time [88, 141].

Latency and cloud access. Analysis of two years of IBM Quantum cloud usage found queueing times ranging from under a minute to over days and a threefold variation in fidelity across machines [139]. A D-Wave anneal takes microseconds [123]. The end-to-end latency, with network access, embedding, and post-processing, is far longer, and robotics studies rarely report it. The one study that did measured 10,110 s for a workflow that a classical solver finished in 0.03 s [175].

Quantum-classical communication and resources. Variational algorithms require many iterations of circuit execution, measurement, and classical update [111], each crossing the quantum-classical boundary. Data encoding costs circuit depth too. HHL-type speed-ups presuppose efficient state preparation and allow only limited read-out [1, 74], and high-dimensional robot observations have to be compressed before encoding [31]. Estimating an expectation value to precision ϵ takes $O(1/\epsilon^2)$ shots, multiplied by the mitigation overhead [25] and by the number of optimizer iterations. On the simulation side, the robotics circuits hit classical memory limits at about 27–40 qubits [4, 30].

Size, weight, power, and cooling. Cryogenic quantum computers cannot be carried by mobile robots [92, 108], and cold-atom sensors still need substantial SWaP reductions [19].

Evidence level. Hybrid architectures are demonstrated at L3, and at L4 only in simulation-coupled settings, such as Webots-Qiskit swarms [108] and CARLA [154]. Two cart-pole studies give the first measured latency budgets. A cloud QPU took seconds per decision where the control loop needed milliseconds [157], and direct programming of the control electronics sped up on-premise execution by more than an order of magnitude without yet reaching closed-loop rates [119].

The latency problem. More and better qubits will not solve this on their own. A shared, cryogenic processor reached over a network cannot sit inside a millisecond control loop, so the constraint is architectural as much as technological. For the near term, the most promising architecture keeps the QPU out of the real-time loop and uses it for training, for periodic fleet-level optimization with a classical fallback, or for sensing.

9 QUANTUM COGNITION FOR ROBOT DECISION-MAKING

9.1 Motivation and Formalism

More and more, robots make decisions with people and about people, and people do not reason by the rules of classical (Bayesian) probability. Question-order effects, conjunction and disjunction fallacies, and violations of the sure-thing principle are well documented [133]. Classical robot

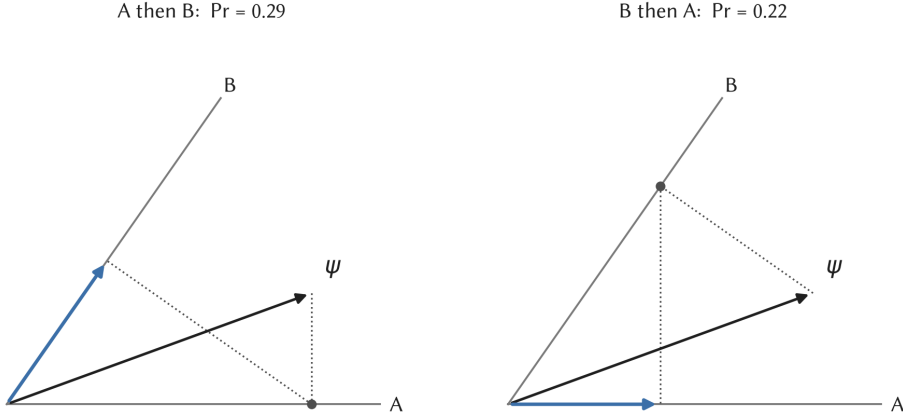


Fig. 8. Why the order of questions matters in quantum probability. A belief state ψ is projected first onto one answer subspace and then onto the other; because the projectors do not commute, the probability of answering yes to both depends on the order. The angles ($\psi = 20^\circ$, $A = 0^\circ$, $B = 55^\circ$) are illustrative. Adapted from the geometric account of Busemeyer and Bruza [23] and Pothos and Busemeyer [133], which Widdows et al. [173] implemented as single-qubit circuits.

decision-making relies on Markov decision processes and POMDPs [83, 95] and on dynamic stochastic models of choice such as decision field theory [24], all of which assume classical, context-independent probabilities. Quantum cognition offers a different formalism, and it needs neither a quantum computer nor a “quantum brain”. In quantum probability (QP), beliefs are unit vectors in a Hilbert space, propositions are subspaces, and judgments are projections [23]. Probabilities are squared lengths of projections, so the order of sequential judgments matters because projectors need not commute,

$$\Pr(A \text{ then } B) = \|P_B P_A \psi\|^2 \neq \|P_A P_B \psi\|^2 = \Pr(B \text{ then } A), \quad (2)$$

and interference terms can make the probability of a disjunction lower than that of either component. These properties reproduce the empirical violations with few parameters [133, 134]. A cognitive state can stay indefinite with respect to a judgment until a decision is required. Composite judgments can be entangled, which represents concept combinations that cannot be decomposed [23]. And contextuality, the dependence of an outcome on the other measurements made with it, explains why the order and framing of information change decisions [134]. This is what makes the formalism interesting for robots. It can model how the order of a robot’s own observations, questions, or recommendations shifts what a human partner is likely to answer, something classical POMDP observation models do not capture. Figure 8 gives the geometric picture behind the question-order effect.

9.2 Robotic and Human-Machine Applications

Widdows et al. [173] showed that several QP models of human decision-making can be implemented directly as circuits on quantum hardware: mental states are represented in qubit registers, and judgments correspond to gates and projective measurements. With reusable circuit components for question-order effects and decisions under uncertainty, experiments on an 11-qubit IonQ trapped-ion computer showed that very small circuits, some with a single qubit, reproduce well-documented

non-classical patterns of human judgment. The authors are careful to claim neither that the brain uses qubits nor any computational advantage, since models this small are easy to simulate classically. What robotics gains is a set of building blocks for putting QP models into a robot's decision module, for example to predict human responses in human-robot interaction (HRI). Yan et al. [177] proposed "quantum affective computing". A robot's emotion is a superposition over basis emotions in a qubit-encoded emotion space, unitary operators model the transitions between emotional states, and the emotions of communicating robots are fused through entanglement, which saves qubits and gates when representing a whole group. It has been shown only in small simulations, not against human emotional data.

De Carolis et al. [38] reviewed more than 35 studies published between 2021 and 2025 on quantum computing for social robotics, as groundwork for a project on quantum-enhanced HRI. They proposed recasting HRI problems such as emotion recognition, path finding, and cognitive modeling as quantum search or QML problems, without yet reporting an implementation. Dağlarlı [37] integrated quantum routines (Deutsch-Jozsa, Bernstein-Vazirani, and Grover search) into a brain-inspired cognitive architecture for autonomous agents that combines a prefrontal-cortex-inspired model, transformer-based RL, metacognition, and situation awareness. In a Minecraft-based simulation, the quantum-enhanced agent reached an 80% success rate about twice as fast as its classical counterpart and collected about 15% more cumulative reward. The circuits were simulated, however, and the contribution of each routine is not isolated.

Quantum-like models have also been proposed for human-machine teams, the setting in which robots increasingly operate. Lawless et al. [99] reviewed a quantum-like theory of interdependence for autonomous human-machine teams, arguing that team members' states are non-separable, analogous to entanglement, so that aggregating individual measurements cannot predict team performance. Humr et al. [78] argued that QP can improve human-AI decision-making by modeling how people rationalize AI outputs: order effects, interference, and "ontic" indecision produce belief dynamics that classical probability cannot capture. Although framed for AI decision support, the approach transfers directly to HRI, where a robot's recommendations shape a human partner's judgments. The only decision-making study set on a vehicle task is Essalmi et al. [51]. Their multi-player quantum game model for automated driving lets each vehicle hold superposed and entangled strategies. In simulated interactive scenarios it raised success rates and reduced collisions compared with classical game-theoretic models, but it has not yet been tested with human drivers or on a real vehicle.

Maturity. The evidence for QP models of human judgment is extensive and experimentally tested [134]. Their robotic use, by contrast, is at L1–L3: small circuits on an 11-qubit trapped-ion computer [173], simulated multi-robot emotion fusion [177], a quantum-enhanced cognitive architecture in a game simulation [37], a quantum game model for simulated driving interactions [51], theoretical models of human-machine teams [99] and human-AI decisions [78], and a project-level agenda for social robots [38].

What is missing. No study has yet tested a QP-based robot decision module against a classical baseline with human participants. There is an irony here. QP models are small and run classically or on current QPUs, so this branch needs less quantum hardware than any other in the survey, and yet it is the least tested on robots. That it could be deployed today remains a hypothesis until someone runs the experiment with people.

10 EMERGING DIRECTIONS

10.1 Quantum-Inspired Robotics

Quantum-inspired algorithms borrow quantum concepts but run on classical hardware. They include quantum-inspired evolutionary algorithms with Q-bit individuals [73], quantum-behaved particle swarm optimization (QPSO), amplitude-based QiRL [45], simulated-bifurcation Ising machines [68], and dequantized sampling-based linear algebra [33, 161]. Zuo et al. [185] improved FastSLAM by using QPSO to move sampled particles closer to the robot's true pose before weighting, with Gaussian perturbations to maintain diversity and a diversity-triggered resampling step to counter particle depletion. In simulation, QPSO-FastSLAM achieved lower pose and landmark errors than FastSLAM 2.0 and PSO-FastSLAM (for example, 0.44 m versus 0.52 m with 20 particles), and it also performed better on the Victoria Park outdoor dataset. Fazilat and Zioui [55] used "quantum neural networks", in fact multilayer perceptrons with a quantum-inspired activation function, to learn the IK of a six-degree-of-freedom ABB IRB140 arm; the networks reduced the mean absolute error by 15.6% (without singularity avoidance) and 16.7% (with it) compared with classical networks, reaching 1.64 mm position accuracy. The paper carries a "quantum" title, but the method runs entirely on classical hardware. It is a good example of why quantum-inspired and genuinely quantum results have to be reported separately. Simulated quantum circuits have also been used for education, for example in Lego robots controlled by multiple-valued quantum automata [137].

Quantum-inspired methods account for most of the quantum-labeled computational results validated on physical robots or real robot data [45, 55, 185], and quantum-inspired Ising machines already solve QUBOs of 2,000 spins on a single FPGA and of 100,000 spins on GPU clusters [68]. They can be deployed today. QiRL has been demonstrated on a physical robot (L5), and QPSO-SLAM has been evaluated on a real outdoor dataset. Dequantization results [33, 161] and the "fine print" of quantum speed-ups [1] show that quantum claims must be compared with the best classical algorithm under the same input assumptions.

10.2 Quantum Swarm Robotics

Quantum swarm robotics has largely been the work of one group in Palermo. Mannone et al. [106] classified swarms with category theory and proposed a block-matrix representation in which diagonal blocks describe individual robots and off-diagonal blocks describe pairwise interactions, computed with a four-qubit circuit on IBM simulators and quantum computers. Mannone et al. [107] implemented local, pairwise interactions as a quantum circuit built around a custom logic gate, embedded it in an ant-foraging algorithm with toy swarms of 3 and 10 robots, and analyzed how global foraging behavior emerges and what role entanglement plays. The authors present this as a way of modeling swarms, not as an optimization method. Mannone et al. [108] reviewed and extended this local-to-global approach, coupling the Webots robot simulator with the Qiskit simulator in two- and three-dimensional search-and-rescue and aquatic scenarios, and reported faster convergence and shorter travel than fully classical algorithms. They are open about the limits: small robots cannot carry quantum processors, the file-based coupling of the two simulators slows everything down, and more realistic sensor-based rewards lower performance. Mannone et al. [105] generalized the block-matrix model to micro- and nano-robotic networks by describing the swarm as a density matrix whose size does not grow with the number of robots. Grover-based path planning [29] and multi-agent QRL with centralized training and decentralized execution [181] complete this direction. The evidence ranges from L1 to L4, with toy swarms of up to 10 robots, and L4 is reached only in simulation [108]. The main barrier is that every robot would need QPU access at every interaction step, and latency and cost grow with the size of the swarm [139].

10.3 Distributed Quantum Robotics

In the distributed vision, fleets or swarms share a pool of networked quantum resources. Quantum communication between mobile robots already has physical demonstrations (Section 7.2). For distributed computation, circuit cutting executes large circuits on several small QPUs with classical recombination [162], multi-agent QRL with centralized training and decentralized execution keeps qubit counts independent of the number of agents [181], and entanglement-based fusion has been proposed to compress the joint state of a robot group [177]. Distributed quantum computation for robots is still at L1–L2. The classical post-processing of circuit cutting grows exponentially with the number of cuts [162], and networked entanglement between mobile platforms is far beyond what can be done today.

10.4 Quantum Computer Vision for Robot Perception

A related literature, usually called quantum computer vision, casts perception problems as QUBOs and solves them with quantum annealing. These studies fail our inclusion criterion I1, because the problems are posed on vision benchmarks and not on robots, so we did not grade them. We discuss them anyway, because several of the problems are front-end components of robot perception and SLAM. Golyanik and Theobalt [64] formulated point-set alignment, the core of scan registration, for an adiabatic quantum computer, and Meli et al. [112] made it iterative so that it reaches high precision with consistent rotation matrices. Birdal et al. [17] introduced the first quantum algorithm for permutation synchronization, which enforces consistent correspondences across multiple views, and Seelbach Benkner et al. [151] solved shape correspondence, posed as a quadratic assignment problem, by iterating small annealer-sized subproblems, which handles problems an order of magnitude larger than earlier quantum formulations. Doan et al. [43] proposed a hybrid quantum-classical algorithm for robust fitting with outliers, and Zaech et al. [182] formulated multi-object tracking, the data-association step of tracking for driving and mobile robots, for adiabatic quantum computers. Evaluated on our hierarchy, these studies would sit at L2–L3: they are executed on annealers or in simulation for small instances and are not coupled to a robot. They share the limits identified for robot optimization in Section 4: problems must be discretized to fit current devices, embedding inflates qubit counts, and mature classical solvers remain faster at practical sizes. Even so, they are the most direct route by which quantum optimization could reach robot perception. Testing one of them inside a registration or tracking pipeline on robot data would be a natural next step, and would bring it into the scope of this survey.

10.5 Quantum Simulation for Robot Materials

Quantum simulation of molecules, the application with the strongest expected advantage [11, 109], may benefit robotics indirectly through better batteries, actuators, and sensors. Variational simulations of lithium-sulfur battery molecules have run on IBM hardware [142], and fault-tolerant resource estimates exist for lithium-ion electrolyte molecules [88] and for catalytic reaction mechanisms [141]. This direction is at L3 for small molecules that remain classically tractable; industrially relevant problems require large fault-tolerant machines.

11 CROSS-CUTTING ANALYSIS

11.1 Synthesis across Robotic Functions

Table 3 brings the results together by robotic function. Each function is served by a different quantum paradigm, and the paradigms are at very different stages. Sensing and communication have been fielded on moving platforms, optimization has reached QPU experiments at fleet scale, and learning and planning are still mostly simulated. One pattern stood out to us: the most mature

“quantum” results in computation are quantum-inspired and run on classical hardware [45, 185]. We did not find a genuinely quantum computational result that is both large enough to be classically hard and better than a properly sized classical baseline [48, 77, 100, 123].

11.2 Evidence Hierarchy

Figure 9 places the 65 empirical and theoretical primary studies (the 7 reviews are excluded) on the evidence hierarchy of Table 2. Most of the weight sits at the bottom: 6 studies are at L1, 19 at L2, 23 at L3, and 10 at L4, all of the latter in simulated robot environments. At L5, quantum communication has been demonstrated on drones, vehicles, and cooperating robots [35, 87, 101, 167], one quantum-inspired learning method has run on a physical robot [45], and one simulated variational policy has balanced a physical cart-pole [157], giving six studies at that level. Only one quantum technology, magnetic-anomaly navigation with quantum magnetometers, has reached end-to-end field deployment with a measured advantage over a classical baseline (L6) [117]. No genuinely quantum computational method has run on a QPU while controlling a physical system; the one that reached L5 executes its circuit in classical simulation [157]. We should also be honest about the single L6 rating. It rests on a company-led preprint, and the airborne trials used a crewed aircraft, not an uncrewed robot [117]. Discount it, and no study in the corpus goes beyond L5. The ranking of the branches would stay the same, because quantum communication would still be fielded on moving robots, but the claim that quantum sensing has improved navigation in the field would then rest on laboratory and simulation evidence alone [28, 170, 171].

Table 4 adds the quantum-execution tag to this picture. Only 22 of the 65 studies ran any part of their quantum computation on a quantum processor, and 21 of them sit at L3. The ten L4 studies are robot-in-the-loop only in simulation, and nine of them also simulate their quantum circuits; the one exception routes subproblems of a simulated 1,000-AGV fleet to a D-Wave annealer [136]. An L4 rating therefore means realistic robot integration, not stronger quantum evidence than L3, and the level and the tag need to be read together. Table 5 looks more closely at the 22 studies that did use a quantum processor. Two things stand out. The instances run on the hardware are small, from one qubit to a few thousand annealer variables, and the large fleets reach the device only as decomposed subproblems [136]. And only six of the 22 compared against a mature classical method, a commercial solver or a state-of-the-art planner [62, 100, 123, 124, 156, 175]. Only two of the six report an edge over the classical side, one in a short paper with a proprietary hybrid solver [124] and one in which the quantum or quantum-inspired solver handles small subproblems inside a classical optimal framework [62]; in neither is the gain attributable to the quantum processor alone.

11.3 Claims of Quantum Advantage versus Evidence

Claims of quantum advantage in the robotics literature refer to four different things, which should not be confused.

Theoretical speed-up. Provable speed-ups relevant to robotics are quadratic for search, amplitude estimation, and Monte Carlo sampling [20, 69, 115], and exponential only for structured problems such as linear systems under strong input/output assumptions [1, 74]. Dequantization has since weakened many of these [33, 161]. For NP-hard optimization there is no provable advantage [52], and a provable learning separation for quantum policies exists only for artificial tasks built on the discrete logarithm [80].

Simulated speed-up. Simulated studies usually report fewer iterations, oracle queries, or parameters, not lower wall-clock time: $O(\sqrt{N})$ query counts for localization [4] and planning [30], faster

Table 3. Cross-domain synthesis of quantum and quantum-inspired methods for robotics. Evidence levels follow Table 2; “sim.” denotes a simulated robot environment.

Application	Quantum method	Evidence	Current limitation	Deployment potential
Fleet and multi-robot routing, AGV and storage-system scheduling [34, 100, 123, 136, 156, 176]	QUBO with quantum annealing or hybrid solvers	L3–L4 (sim.)	Decomposition needed; classical and digital annealers competitive or better	Medium: periodic, offline
Multi-agent pathfinding [62, 66]	QUBO subproblems in exact classical framework	L2–L3	Small QUBOs; structure-dependent	Medium
Industrial trajectories, assembly, inspection [124, 148, 175]	QUBO and annealing in classical pipelines	L3	Classical algorithm does most of the work; overheads	Low-medium
Warehouse picking [6]	QUBO with VQE and QAOA via cloud	L3	Tiny instances; cloud latency	Low-medium
Inverse kinematics [122, 126, 146]	Gate-based VQA, annealing, QML with Grover	L2–L3	Toy size; continuous classical IK solvers stronger	Low
Motion and swarm planning [14, 29, 30, 97]	Grover search	L1–L2	Oracle depth; noiseless simulation	Low (needs fault tolerance)
UAV path, routing, and task assignment; view planning [9, 79, 125, 147, 180]	QAOA, annealing, variational, hybrid quantum ACO	L2–L3	Small graphs; weak baselines	Low-medium
Localization [4]	Grover pattern search	L3	Noise; simulation limited to about 40 qubits	Low (needs fault tolerance)
Navigation, locomotion, and multi-agent RL [3, 48, 77, 103, 129, 130, 159]	VQC-based deep RL	L3–L4 (sim.)	No advantage over adequately sized classical networks	Medium: compact embedded policies
Driving and arm control [2, 154]	Quantum critic; variational SAC	L4 (sim.)	Simulation only	Medium (training-time use)
Real-time control (cart-pole) [119, 157]	Variational policies; QPU inference	L3–L5	Toy system; cloud QPU too slow; simulated circuit on hardware	Medium (shows latency limits)
Object detection for driving [145]	Quantum CNN	L2	Simulated circuits	Low-medium
Navigation, SLAM, IK (quantum-inspired) [45, 55, 185]	QiRL, QPSO, quantum-inspired networks	L2–L5	Weak baselines	High (runs classically today)
GNSS-denied navigation [28, 117, 164]	Quantum magnetometers, atom interferometers	L3–L6	SWaP-C, bandwidth	High (near-term)
Human decision modeling, HRI, human-machine teams, driving interaction [51, 78, 99, 173, 177]	Quantum-probability models	L1–L3	No HRI evaluation yet	Medium-high (hardware-light; untested on robots)
Swarms and distributed robots [107, 108]	Pairwise quantum circuits	L3–L4 (sim.)	QPU access at every interaction	Low
Secure robot and drone communication [35, 87, 101, 167]	QKD, photonic hardware	L5	Range, SWaP, weather	High (security, near-term)

Table 4. Evidence level (rows) by where the quantum component was executed (columns) for the 65 primary studies (7 reviews excluded). “Simulated” includes two studies of simulated quantum sensors [170, 171]; HW = hardware; QI = quantum-inspired algorithms on conventional hardware.

Level	None	Simulated	QPU	Sensor/link HW	Classical (QI)	Total
L1 Theoretical	6	–	–	–	–	6
L2 Simulation	–	17	–	–	2	19
L3 QPU experiment	–	–	21	2	–	23
L4 Robot-in-the-loop	–	9	1	–	–	10
L5 Physical demonstration	–	1	–	4	1	6
L6 End-to-end deployment	–	–	–	1	–	1
Total	6	27	22	7	3	65

Table 5. The 22 primary studies that ran part of their quantum computation on a quantum processor: device, the largest instance run on it, the strongest classical baseline, and the outcome. – = no classical comparison recorded in our extraction; DA = Fujitsu Digital Annealer; MCL = Monte Carlo localization.

Study	Device	Largest instance on the QPU	Strongest baseline	Outcome
Clark et al. [34]	D-Wave 2000Q	Up to 200 robots on a grid	Classical benchmark runs	Valid plans; anneal time only
Ohzeki et al. [123]	D-Wave 2000Q	10 AGVs, ≤ 60 variables	Gurobi; Digital Annealer	Classical and DA handle larger instances
Quang et al. [136]	D-Wave	Sub-QUBOs ($< 10,000$ variables) of a 1,000-AGV fleet	–	No quantum-vs-classical comparison
Śmierzchalski et al. [156]	D-Wave hybrid	21 AGVs ($\approx 1,300$ variables)	CPLEX	Comparable
Leib et al. [100]	D-Wave hybrid	Laboratory transport scheduling	Gurobi; Digital Annealer	Gurobi 10–100 $\times$ faster on small; DA best on largest
Schuetz et al. [148]	D-Wave (Braket)	Reduced sub-problems	Classical BRKGA	Performance from classical algorithm
Willmann et al. [175]	D-Wave Advantage 4.1	4 tasks, 2 workstations	Exact classical solver	10,110 s vs. 0.03 s end to end
Osaba et al. [124]	D-Wave hybrid	5 inspection cases	Gurobi; OR-Tools	Competitive quality, less time (preliminary)
Atchade-Adelomou et al. [6]	IBM, Braket, D-Wave	Small warehouse instances	–	Valid routes
Windmann [176]	IBM Q System One	Small AS/RS instances	–	Only small instances fit
Salloum et al. [146]	D-Wave Pegasus, Zephyr	Two-link planar IK	Classical QUBO solvers	Lower time-to-solution on larger instances; not better than continuous IK
Otani et al. [126]	RIKEN/Fujitsu 64-qubit	2 qubits	–	Entanglement helps vs. no-entanglement circuit
Gonçalves [65]	IBM 5-qubit	A few qubits	–	Feasibility only
Gerlach et al. [62]	D-Wave Advantage	Small QUBO subproblems in BCP	BCP, EECBS, LaCAM*, LNS2	Competitive with state of the art
Innan et al. [79]	IBM ibm_kyiv	Small flight graphs	–	Feasible paths despite noise
Osaba et al. [125]	D-Wave	3 drone-routing cases	–	Feasible routes
Antero et al. [4]	IBM Brisbane (127-qubit); IonQ noise model	Small 2D grid maps	MCL (analytical)	Accuracy falls as maps grow
Altrabulsi et al. [3]	IBM ibm_fez	Trained 8-qubit policy	Tabular Q, MLP, SNN (sim.)	Hardware run as feasibility check
Ngo et al. [119]	VTT Q5	1-qubit agent	–	$> 10\times$ faster execution, below control rate
Widdows et al. [173]	IonQ 11-qubit	1 to a few qubits	n/a	Reproduces judgment patterns
Mannone et al. [106]	IBM	4-qubit swarm model	–	Toy model
Mannone et al. [107]	IBM	Swarms of 3 and 10 robots	–	Modeling only

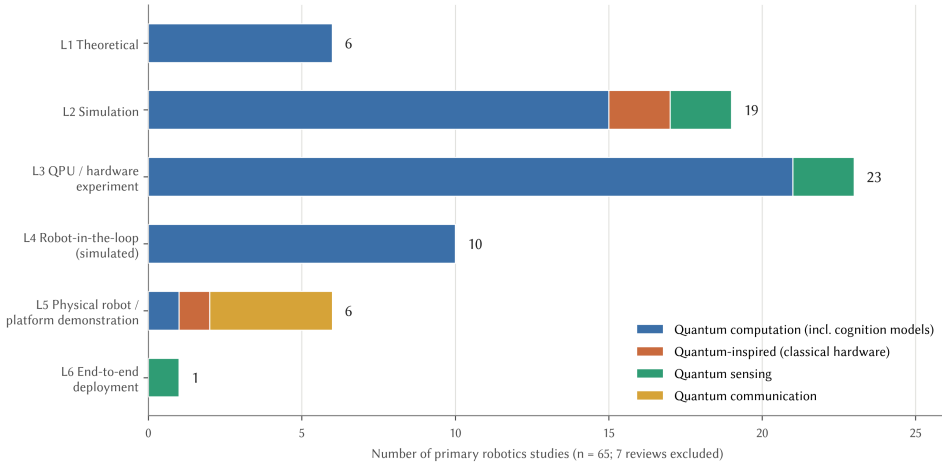


Fig. 9. Evidence levels reached by the 65 primary robotics studies (7 reviews excluded), by technology type. The only quantum computational study at L5 runs its circuit in classical simulation on a physical cart-pole; the other studies at L5 and L6 use quantum sensing, quantum communication, or quantum-inspired classical computation.

swarm convergence against generic ant simulations [29], and parameter reductions of about 100–200 times in QRL [2, 48]. None of these metrics accounts for circuit depth, shots, error mitigation, or the cost of classical simulation, and the baselines are rarely state of the art.

Hardware demonstration. Hardware demonstrations show that something can be done, not that it is better. The list is longer than one might expect: valid annealer plans for up to 200 robots [34], collision-free AGV control [123], decomposed routing of a 1,000-AGV fleet [136], optimal MAPF with QUBO subproblems [62], QAOA drone paths on IBM hardware [79], annealer-based IK [146], two-qubit kinematics on a 64-qubit processor [126], storage-and-retrieval scheduling with QAOA [176], drone routing on annealers [125], a single-qubit control agent on a superconducting QPU [119], a navigation policy executed on IBM hardware [3], and cognitive circuits on trapped ions [173]. Wherever a comparison was made, classical solvers stayed competitive or did better [100, 123, 156, 175]. The advantages on the largest instances came from quantum-inspired digital hardware [100], and hardware noise cost accuracy [4].

End-to-end robotics advantage. Better task performance in a realistic robotic deployment against a strong classical baseline has been shown only for quantum sensing [117]. Quantum communication has been deployed between moving platforms [35], but it offers security rather than a performance advantage. For quantum computation, nobody has yet shown an end-to-end advantage on a robot.

11.4 The Latency Gap

Figure 10 compares the relevant timescales. Robot control loops run at 10–1,000 Hz and planning loops at roughly 0.1–10 Hz, whereas cloud QPU access involves queueing from under a minute to several days [139], on top of compilation, embedding, many shots, and post-processing. Variational methods multiply all of this by hundreds of optimizer iterations [111]. That leaves only three ways to integrate a QPU today: offline training [154], asynchronous periodic optimization with a classical fallback [123], and sensing [117]. Dynamic circuits [36] and on-premise edge QPUs may shorten latency in the future, but real-time quantum control of robots is not around the corner.

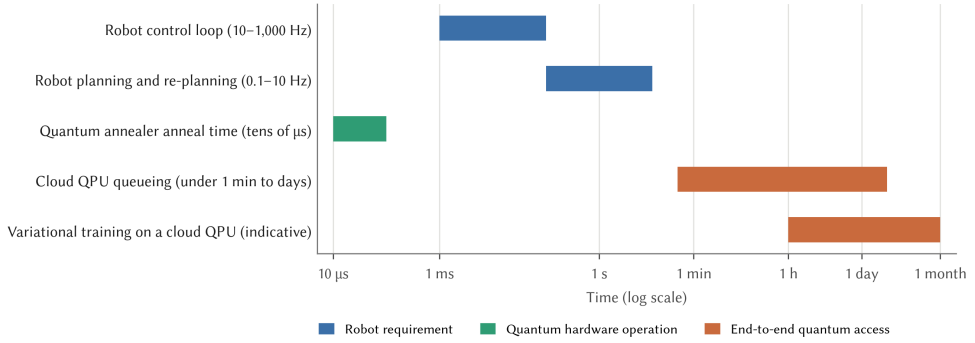


Fig. 10. Timescales of robot control and planning compared with quantum hardware operation and end-to-end cloud quantum access (log scale). The anneal time is from Ohzeki et al. [123] and the queueing range from Ravi et al. [139]; the range for variational training is indicative.

11.5 Scalability

Scaling runs into limits on four fronts. QAOA needs hundreds to thousands of qubits before any advantage appears [70]. Barren plateaus limit trainability [96, 110]. Robot sensor data must be compressed before it can be encoded [31]. And results become hard to verify, because classical simulation of the circuits used in robotics stops at about 27–40 qubits [4, 30]. So far the robotics studies have stayed at toy scale: grid worlds, swarms of up to 10 robots, two-link manipulators. The hardware adds its own limits: noise and decoherence [135], exponential error-mitigation overheads [25], fault-tolerance overheads of hundreds to about a thousand physical qubits per logical qubit [21, 58], cryogenic SWaP-C [92], restricted annealer connectivity, and the bandwidth, dynamic-range, and size limits of quantum sensors [19, 170].

11.6 A Worked Resource Estimate

Several studies in this survey rest their case on a quadratic speed-up that would need fault-tolerant hardware. To see what such a speed-up would buy a robot, we made a rough estimate for the clearest example, Grover-based localization on a known map [4]. The numbers are ours and deliberately generous to the quantum side; Table 6 lists the assumptions.

Take a $1,000 \times 1,000$ occupancy grid, $N = 10^6$ cells, which at 5 cm resolution covers a 50×50 m floor. Grover search needs about $\frac{\pi}{4}\sqrt{N} \approx 785$ iterations. Each iteration calls the oracle, and the oracle must read the map: without quantum random-access memory, which does not exist as hardware, the standard way to load N classical entries into a circuit costs about $4N$ T gates, or roughly N Toffoli gates, per call [7], or about $2N$ when the call is also uncomputed. At $10 \mu\text{s}$ per error-corrected Toffoli gate, about 17 times faster than the $170 \mu\text{s}$ that Babbush et al. [8] estimate for a surface-code machine, the search takes about $785 \times 2 \times 10^6 \times 10 \mu\text{s} \approx 1.6 \times 10^4 \text{ s}$, roughly four and a half hours. A classical scan of the same map, at 10 ns per cell, takes 10 ms. The quantum search therefore loses by six orders of magnitude, because loading the map costs as much as the classical search it was meant to replace.

Suppose instead that a qRAM with logarithmic access time existed, so that each oracle call cost only 10^3 Toffoli gates. The search would then take about 16 s, still far slower than the 10 ms classical scan and far outside a localization loop that runs at several hertz. Under these assumptions the two methods break even only for maps of about 2.5×10^{12} cells, where each search takes about seven hours. The machine itself would need on the order of 170 logical qubits for the address, pattern, and

Table 6. Assumptions and results of the resource estimate for Grover localization (Section 11.6). All values are order-of-magnitude estimates chosen to favor the quantum method.

Quantity	Value
Map size N	10^6 cells ($1,000 \times 1,000$ grid)
Grover iterations	$\frac{\pi}{4} \sqrt{N} \approx 785$
Oracle cost, map loaded by table lookup	$\approx 2N$ Toffoli gates per iteration [7]
Oracle cost, hypothetical log-time qRAM	$\approx 10^3$ Toffoli gates per iteration
Error-corrected Toffoli time	$10 \mu\text{s}$ (vs. $\approx 170 \mu\text{s}$ estimated by Babbush et al. [8])
Classical scan	10 ns per cell, 10 ms per map
Quantum time, table lookup	$\approx 1.6 \times 10^4 \text{ s}$ (about 4.4 h)
Quantum time, with qRAM	$\approx 16 \text{ s}$
Break-even map size, with qRAM	$\approx 2.5 \times 10^{12}$ cells (about 7 h per search)
Logical / physical qubits	$\approx 170 / \approx 2 \times 10^5$ (distance 25), excluding factories

comparison registers, or a few hundred thousand physical qubits at surface-code distance 25 [58], before counting the magic-state factories. These figures agree with the general analysis of Babbush et al. [8], who argue that quadratic speed-ups are unlikely to give a runtime advantage on early fault-tolerant machines. The estimate is crude, and better circuits would shift the numbers, but not by the six orders of magnitude needed to change the conclusion.

12 OPEN CHALLENGES AND RESEARCH AGENDA

We take the open challenges from the taxonomy of Figure 3. For each robotic function we describe the gap between today's evidence and a deployable result, and then turn to the challenges that cut across all functions. Table 8 arranges the resulting research directions by time horizon.

12.1 Challenges by Robotic Function

Optimization. The open question is whether quantum hardware adds anything to a well-engineered QUBO pipeline. Studies need *instance families* on which quantum, quantum-inspired, and classical solvers are compared on equal end-to-end terms, including embedding and queueing, at the sizes of real fleets [100, 136]. Constraint-preserving mixers [71] have been tested on robot task allocation only in simulation [9], and transferable QAOA parameters [183] not at all. The most promising template, in our view, places small QUBOs inside a provably optimal classical framework [62].

Planning. Grover-based planning and localization need oracle constructions whose qubit counts and depth scale gracefully with map size, and evaluations against A^* , RRT* [86], and Monte Carlo localization [40] on realistic maps. Our rough estimate in Section 11.6 suggests that, for localization at least, the quadratic speed-up does not survive the cost of loading the map into the oracle; a careful estimate by specialists, with optimized circuits, would settle the question for planning as well [8]. A useful first experiment is modest: take one published Grover planner, count its logical resources exactly, and compare the resulting wall-clock time with A^* on the same map.

Learning. The parameter-efficiency hypothesis of QRL [2, 77] should be tested for what matters on robots: sample efficiency, training wall-clock time, energy, and robustness on physical platforms, with circuits trained on real QPUs rather than in idealized simulation. Encodings for high-dimensional observations that avoid barren plateaus [27, 96] without becoming classically

simulable remain open. The cleanest test would fix a robot task and a sample budget, train a quantum and an adequately sized classical policy under the same budget, and report sample efficiency and wall-clock time; a parameter count alone should no longer be accepted as the headline result.

Sensing and communication. Quantum navigation sensors need independent field replication [117], miniaturization to the SWaP budgets of small robots [19], and integration into standard estimation stacks through Kalman-type fusion [28, 170]. QKD between robots needs evaluations of range, weather, and alignment under realistic mission profiles [35]. The decisive experiment for navigation is a GNSS-denied mission flown by an uncrewed robot, with the quantum fix inside its own estimator and the same mission flown with a classical INS for comparison; for communication, it is a coordination task whose commands are authenticated with quantum-distributed keys.

Decision-making. The clearest near-term gap is a QP-based decision and human-modeling layer for robots, evaluated with human participants against Bayesian and POMDP baselines [95, 134]. Such a layer needs little or no quantum hardware [173] and addresses a genuine need in HRI and human-machine teaming [78, 99]. A first study could be small: a robot that asks two questions in either order, a QP model and a Bayesian model fitted to the same participants, and a comparison of how well each predicts the answers to held-out question orders.

Where we would place our bets. If we had to choose, we would invest first in quantum navigation sensing on uncrewed platforms and in QP models for HRI, because both can be tested now and neither depends on fault-tolerant hardware. Next comes fleet-level optimization, but only in studies that report end-to-end time against the best classical solver, since that is where the case for quantum hardware is currently weakest. We would treat Grover-based planning and real-time quantum control as long-term research, worth pursuing for what they teach about algorithm design but not yet as routes to a better robot.

12.2 Cross-Cutting Challenges

Reporting standards and benchmarks. We would have liked to pool the results quantitatively, but the tasks, metrics, and baselines differ too much. Table 7 proposes a minimal reporting checklist for quantum-robotics experiments; shared benchmark suites for fleet scheduling, MAPF, navigation RL, and localization would allow results to be compared across studies. Application-level benchmark frameworks already exist outside robotics. QUARK, for example, includes a robot-path reference problem with quantum and classical solvers [57], and extending such frameworks to the tasks in this survey would be a direct way to meet the checklist.

Separating quantum from quantum-inspired results. Quantum-inspired and classical results should be reported separately from quantum-hardware results, and claimed speed-ups should be checked against the best classical algorithm under the same input assumptions [1, 161].

Latency-aware architectures. Architectures should keep the QPU out of the real-time loop, provide classical fallbacks, and report end-to-end latency budgets. Edge QPUs with dynamic circuits [36] and circuit cutting [162] are the main technical routes to closing the latency gap.

Safety and verification. A robot should not act on a probabilistic quantum output without checking it first. Post-processing that rejects constraint-violating samples [123] is a start, but there are no formal guarantees yet for hybrid quantum-classical controllers.

13 CONCLUSION

We set out to find what the published evidence says about quantum technologies for robots. After reading 72 primary studies against 107 foundational works, our answer is that quantum robotics is

Table 7. Proposed minimal reporting checklist for quantum-robotics experiments.

Item	What to report
1. Task and formulation	Robotic task, its mapping to the quantum formulation (e.g., QUBO size, penalty weights, encoding), and any discretization relative to the original continuous problem.
2. Quantum type	Genuinely quantum or quantum-inspired; quantum hardware, simulator, or digital annealer used; noise model for simulations.
3. Classical baseline	State-of-the-art, tuned classical method (e.g., Gurobi, CPLEX, RRT*, A*, adequately sized deep RL) with comparable resources.
4. Quantum resources	Qubits, circuit depth, two-qubit gate count, shots, optimizer iterations, embedding size, and error-mitigation method.
5. End-to-end time and energy	Wall-clock time including queueing, compilation or embedding, and post-processing; energy where possible.
6. Task metrics	Success rate, solution quality, path length, or error, with variance over seeds or instances.
7. Evidence level	Level reached in Table 2, and whether the quantum component runs in the real-time loop.
8. Reproducibility	Code, data, hyperparameters, device name, and calibration date.

Table 8. Research agenda derived from the taxonomy, arranged by time horizon.

Horizon	Research direction	Function	Key references
Near term (1–3 years)	Integrate quantum magnetometers and cold-atom accelerometers into robot navigation stacks with Kalman-type fusion and independent field validation	Sensing	[28, 117]
	Develop QP decision and human-modeling modules for social and collaborative robots; evaluate them with human participants against Bayesian and POMDP baselines	Decision-making	[95, 134]
	Establish benchmarks and reporting standards (Table 7)	All	-
	Use quantum and quantum-inspired QUBO solvers for offline and periodic fleet optimization with classical fallbacks	Optimization	[62, 68, 136]
	Evaluate QKD for secure multi-robot and drone communication	Communication	[35]
Medium term (3–5 years)	Train QRL policies on real QPUs where parameter efficiency matters for embedded deployment, and test for sample-efficiency gains on physical robots	Learning	[77, 93]
	Build edge (on-premise) QPU architectures for warehouses and factories with dynamic circuits and asynchronous execution	Architecture	[36]
	Apply constraint-preserving QAOA mixers with transferable parameters to task allocation and scheduling	Optimization	[71, 183]
	Use fault-tolerant amplitude estimation to accelerate Monte Carlo localization and belief-space planning	Planning	[115]
Long term (5+ years)	Develop distributed quantum computing for multi-robot systems	Architecture	[162, 181]
	Use quantum simulation to design robot batteries and materials	Materials	[88, 142]

rich in ideas but still early in evidence. Quantum optimization is the most hardware-tested area, from 200-robot routing on annealers to decomposed routing of 1,000-AGV fleets, but classical and quantum-inspired solvers match or beat it in head-to-head comparisons. Quantum search and planning remain simulated. Quantum reinforcement learning reliably reduces parameter counts,

but nobody has shown that it produces better or faster policies, and supervised QML faces barren plateaus and dequantization. Quantum sensing is a different story: one field trial, still awaiting independent replication, reports large navigation gains, and QKD already links drones and vehicles in motion. Among the computational results, the ones that have been validated physically are mostly quantum-inspired and run on classical hardware.

For the near term (2026–2030), the realistic opportunities are quantum-enhanced navigation sensing, QP decision models for HRI, offline or periodic QUBO-based fleet optimization with classical fallbacks, and QPU-trained compact policies for embedded deployment. Real-time control will stay classical for the foreseeable future. Until on-premise, low-latency, fault-tolerant processors exist, quantum resources belong outside the millisecond loop, as training accelerators, periodic planners, or sensors. The long-term vision is a hybrid robot in which classical controllers handle real-time control, quantum sensors provide drift-free navigation, remote or edge quantum processors solve fleet-level optimization and train compact policies, and QP models help the robot understand the humans it works with. Getting there will depend less on a single breakthrough than on patient, honest evaluation of each component, on real robots, against the best classical alternative.

DATA AVAILABILITY

The data-extraction table for the 72 primary studies, with evidence level, quantum-execution tag, and publication status for each study, the appraisal criteria, the PRISMA 2020 flow diagram and checklist, and the search strings are available as supplementary material and in an archived dataset [dataset DOI].

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